Quantum-Inspired Time Series Forecasting: A Systematic Review and Neuro-Symbolic Emulation Framework

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ABSTRACT Quantum Machine Learning (QML) has been explored for Time Series Forecasting (TSF) through quantum feature maps, variational circuits, kernel methods, and reservoir-like models. However, its practical use remains limited by the exponential cost of state-vector simulation, the noise and connectivity constraints of NISQ hardware, and the lack of standardized benchmarking against strong classical baselines.

This paper makes two contributions. First, it presents a PRISMA-compliant systematic review of 45 QML-TSF studies published or made publicly available between 2014 and 2025, organized by algorithmic family, encoding strategy, execution modality, evaluation protocol, and reproducibility level. Second, it evaluates a bounded proof-of-concept framework in which a classical neuro-symbolic Cellular Neural Network (CeNN) is used as a functional emulator of selected behaviors of a Quantum Extreme Learning Machine (QELM) reference model.

The proposed CeNN is not claimed to be a quantum computer, a physical state-vector simulator, or evidence of quantum advantage. It is assessed as a classical emulator through three criteria: forecasting utility, functional emulation fidelity, and computational scalability. Emulation is measured using Pearson correlation, KL divergence, spectral distance, autocorrelation-profile distance, and the mean-squared error between CeNN emulation outputs and enriched QELM observables ($(\langle Z_i\rangle, \langle Z_i Z_j\rangle)$). Forecasting performance is evaluated primarily using MASE, with RMSE, MAE, and R^2 as supporting metrics.

Experiments on energy load, financial log-returns, weather temperature, and Mackey–Glass chaotic dynamics show that CeNN achieves competitive forecasting performance against MLP, LSTM, Transformer, naive baselines, and the QELM teacher, but is not uniformly superior across all datasets. Functional emulation is strongest on the Mackey–Glass benchmark and weaker on energy and weather series, revealing a trade-off between forecasting accuracy and QELM alignment. Scalability results support near-linear architectural growth in CeNN parameters, while timing and latency remain hardware-sensitive. These results position CeNN-based functional emulation as a tractable research direction for scalable quantum-inspired TSF, with broader validation left for future work.

INDEX TERMS Quantum machine learning, time series forecasting, quantum-inspired functional emulation, Quantum Extreme Learning Machine, Cellular Neural Networks, neuro-symbolic architectures, functional emulation fidelity, computational scalability, PRISMA systematic review, reproducible benchmarking.

I. INTRODUCTION

Time series forecasting (TSF) is a central task in energy, finance, climate monitoring, telecommunications, transportation, and industrial systems. These applications often involve non-linear dynamics, non-stationarity, long-range temporal dependencies, and noise, which require forecasting models that are both accurate and computationally efficient [1], [2].

Classical statistical models remain important because of their interpretability and theoretical maturity. Machine learning and deep learning methods, including support vector regression, recurrent neural networks, Long Short-Term Memory (LSTM) networks [3], and attention-based Transformers [4], have further improved forecasting performance on complex temporal data. However, these models may still be affected by high training cost, sensitivity to hyperparameters, limited interpretability, and reduced robustness under regime shifts or long-horizon forecasting constraints.

Quantum Machine Learning (QML) has therefore been explored as an alternative representation paradigm for TSF. Quantum feature maps, variational quantum circuits, quantum kernels, Quantum Extreme Learning Machines (QELMs), and quantum reservoir-like models have been investigated as ways of mapping temporal inputs into expressive high-dimensional representations [5]–[9]. Recent work has applied these ideas to energy, finance, climate, traffic, and chaotic systems forecasting [10]–[13].

Despite this promise, the practical status of QML for TSF remains uncertain. Existing studies differ substantially in datasets, forecasting horizons, quantum encodings, baselines, evaluation metrics, and execution backends. Moreover, many reported results rely on small-scale state-vector simulations, while current Noisy Intermediate-Scale Quantum (NISQ) hardware remains constrained by decoherence, gate errors, limited connectivity, and restricted circuit depth [14], [15]. This creates a methodological gap between the representational potential of QML models and their reproducible evaluation on realistic forecasting tasks.

A central distinction in this paper is the difference between quantum simulation and quantum-inspired functional emulation. Quantum simulation aims to reproduce the state evolution of a quantum system, typically through explicit state-vector or density-matrix representations, leading to exponential memory growth in the number of qubits. By contrast, quantum-inspired functional emulation refers here to a classical approximation of selected task-relevant behaviors of a quantum-inspired reference model, such as output distributions, spectral structure, temporal correlations, and observable responses. It does not imply quantum-state equivalence, genuine entanglement reproduction, or quantum computational advantage.

This paper uses the systematic review not only as a descriptive component, but also as a motivation for a bounded proof of concept. The review identifies recurring limitations in QML-TSF research: scalability barriers, heterogeneous benchmarks, limited reproducibility, and weak separation between simulation, hardware execution, and functional emulation. Motivated by these gaps, we investigate whether a classical neuro-symbolic Cellular Neural Network (CeNN) can serve as a scalable functional emulator of selected behaviors of a QELM reference model.

CeNNs are locally coupled dynamical systems originally introduced for parallel distributed processing and spatio-temporal dynamics modeling [16]. In this work, they are not treated as quantum systems. Instead, they are used as classical dynamical substrates whose local connectivity, forecasting head, emulation head, and regularization terms are evaluated for three purposes: forecasting utility, functional alignment with QELM observables, and architectural scalability.

A. SCIENTIFIC HYPOTHESIS AND SCOPE

The central hypothesis of this work is that a neuro-symbolic CeNN-based emulator can approximate selected task-relevant statistical, spectral, and temporal-correlation behaviors associated with a QELM reference model while avoiding the exponential memory scaling of full state-vector simulation. This hypothesis is formulated at the level of functional emulation rather than physical quantum equivalence.

Accordingly, the proposed CeNN framework is not presented as a quantum computer, a physical simulator of quantum states, or evidence of quantum computational advantage. It is evaluated as a classical functional emulator through three criteria: forecasting performance, functional emulation fidelity, and architectural scalability.

The scope of the paper is deliberately bounded. The systematic review identifies methodological, scalability, and reproducibility gaps in QML-based time-series forecasting. The CeNN framework is then evaluated as one possible classical response to these gaps through a proof-of-concept study on four datasets: energy load, financial log-returns, weather temperature, and deterministic chaotic dynamics. Claims about generality, operational deployment, and hardware-level quantum advantage are outside the scope of this work.

B. CONTRIBUTIONS OF THIS WORK

The contributions of this work are limited to three elements:

1) **Systematic review and taxonomy of QML-TSF research:** We conduct a PRISMA-compliant systematic review of QML methods for time series forecasting over the period 2014–2025 [17]. The review organizes the literature according to algorithmic

families, encoding strategies, execution modalities, evaluation protocols, and reproducibility practices.
2) **Formal positioning of quantum-inspired functional emulation:** We clarify the distinction between quantum simulation and quantum-inspired functional emulation. In this paper, functional emulation refers to the classical approximation of selected task-relevant output distributions, spectral structures, temporal correlations, and QELM observable responses, without claiming quantum-state equivalence or quantum computational advantage.
3) **Bounded proof of concept of a CeNN-based emulator:** We evaluate a neuro-symbolic CeNN framework against naive baselines, MLP, LSTM, Transformer, and a QELM reference model. The benchmark assesses forecasting utility, functional emulation fidelity using enriched QELM observables ($\langle Z_i \rangle, \langle Z_i Z_j \rangle$), ablation behavior, and architectural scalability.

C. POSITIONING RELATIVE TO EXISTING REVIEWS

Foundational reviews by Schuld *et al.* [5], Biamonte *et al.* [6], and Dunjko and Briegel [7] established the general foundations of QML. Other works have examined specific families such as variational quantum algorithms [14], quantum reservoir computing [18], [19], and quantum models for forecasting [10], [12]. However, existing reviews do not jointly analyse QML-TSF in terms of scalability, benchmarking fairness, execution modality, reproducibility, and functional emulation.

This paper addresses that gap by linking the systematic review to a bounded methodological proof of concept. The review identifies the limitations that motivate the CeNN study, while the experimental component evaluates whether a locally connected classical dynamical architecture can approximate selected QELM behaviors in a scalable forecasting setting.

D. RESEARCH QUESTIONS

The study is guided by the following research questions:
1) **RQ1:** What algorithmic families, encoding strategies, execution modalities, and evaluation practices characterize the current literature on QML for time series forecasting?
2) **RQ2:** What scalability, benchmarking, and reproducibility limitations prevent current QML-TSF approaches from being reliably extended to larger forecasting settings?
3) **RQ3:** Can a classical neuro-symbolic CeNN be formally positioned as a quantum-inspired functional emulator by targeting selected statistical, spectral, temporal-correlation, and QELM observable behaviors rather than quantum-state fidelity?
4) **RQ4:** Under this functional-emulation framing, does the proposed CeNN proof of concept provide competitive forecasting performance, measurable QELM alignment, and near-linear architectural scaling on selected benchmark datasets?

E. PAPER ORGANIZATION

The remainder of this paper is organized as follows. Section II introduces the foundations of TSF and QML. Section III presents the PRISMA-compliant review methodology. Section IV synthesizes the reviewed QML-TSF literature. Section V defines the CeNN-based functional-emulation framework. Section VI reports the proof-of-concept experiments and results. Section VII discusses methodological gaps and open challenges. Section VIII presents limitations and future hardware-aware validation directions. Section IX reports reproducibility and implementation details. Section X outlines future research directions. Finally, Section XI concludes the paper.

II. FOUNDATIONS OF QUANTUM MACHINE LEARNING FOR TIME SERIES

A. FUNDAMENTALS OF TIME SERIES FORECASTING

Time series forecasting (TSF) aims to estimate future values of a temporal process from its past observations. Given an observed sequence $\{x_t\}_{t=1}^{T}$, the forecasting task consists of predicting $\{x_{T+1}, \ldots, x_{T+h}\}$ over a forecast horizon h. TSF is central to energy systems, finance, telecommunications, meteorology, transportation, and industrial monitoring [1], [2], [12], [20].

Real-world time series often contain trend, seasonality, cyclical behavior, stochastic fluctuations, missing values, regime shifts, and exogenous disturbances. These properties make forecasting difficult, especially when models must remain accurate, robust, and computationally efficient under non-stationary conditions.

Classical TSF methods can be grouped into three broad families. Statistical models, such as ARIMA, SARIMA, and state-space models, provide interpretable and theoretically grounded baselines, but they are often limited under high-dimensional or strongly non-linear dynamics. Machine learning models, including support vector regression, tree-based ensembles, and kernel methods, can capture non-linear relationships but frequently depend on careful feature engineering. Deep learning models, including recurrent neural networks, Long Short-Term Memory (LSTM) networks [3], and Transformers [4], can learn complex sequential patterns, but they may require large datasets, substantial computation, and extensive hyperparameter tuning.

1) Motivation for Quantum and Quantum-Inspired Approaches

Although classical and deep learning models remain dominant in practical forecasting, several limitations motivate the exploration of QML and quantum-inspired approaches. These include the difficulty of compactly representing complex non-linear temporal dependencies, the computational cost of training large neural architectures, sensitivity to noisy and non-stationary data, and limited robustness under distribution shifts.

QML has been proposed as a possible representation paradigm because quantum feature maps, quantum kernels, variational circuits, and reservoir-like quantum dynamics can map classical inputs into high-dimensional Hilbert spaces [5]–[8]. However, the use of quantum representations in forecasting does not automatically imply practical quantum advantage. Current QML-TSF research remains constrained by small-scale simulations, NISQ hardware noise, heterogeneous evaluation protocols, and limited reproducibility [12], [14], [15]. This motivates a careful distinction between physical quantum execution, classical quantum simulation, and quantum-inspired functional emulation.

B. QUANTUM COMPUTING CONCEPTS RELEVANT TO FORECASTING

A qubit is the fundamental unit of quantum information. A pure one-qubit state can be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1, \tag{1}$$

where α and β are complex probability amplitudes. Multi-qubit systems can exhibit entanglement, meaning that the joint state cannot be factorized into independent subsystem states. A standard example is the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right). \tag{2}$$

These concepts are central to quantum information theory. However, the CeNN framework proposed in this paper does not reproduce genuine quantum entanglement or simulate quantum states.

In hybrid QML, classical data are encoded into quantum states through a mapping $U_\phi(x)$, processed by a quantum circuit or quantum dynamical model, and measured to produce classical outputs. For TSF, the input is typically a temporal window

$$w_t = [x_{t-L+1}, \ldots, x_t], \tag{3}$$

which is encoded through angle encoding, amplitude encoding, basis encoding, or problem-specific quantum feature maps. The measured expectation values, such as $\langle Z \rangle$, are then used for forecasting or as features for a classical readout.

C. SIMULATION, HARDWARE EXECUTION, AND FUNCTIONAL EMULATION

A central methodological distinction in this paper is the difference between *quantum simulation*, *physical quantum execution*, and *quantum-inspired functional emulation*.

Quantum simulation refers to the classical numerical representation of a quantum system. For an n-qubit pure state, this usually requires storing a 2^n-dimensional complex state vector. For mixed states, a density matrix may require $2^n \times 2^n$ complex entries. This provides state-level fidelity but leads to exponential memory growth with the number of qubits [14].

TABLE 1. Quantum simulation, physical execution, and functional emulation.

Approach	Target representation	Main limitation
Quantum simulation	State vector or density matrix on classical hardware	Exponential memory and time cost
Physical quantum execution	Quantum circuits on physical qubits	NISQ noise, decoherence, and limited connectivity
Functional emulation	Selected task-relevant input–output behaviors	No physical quantum-state equivalence

Physical quantum execution refers to the direct implementation of quantum circuits or quantum dynamical processes on a quantum processing unit. In the NISQ era, this execution is limited by decoherence, finite coherence times, gate errors, measurement noise, and restricted qubit connectivity [14], [15].

Quantum-inspired functional emulation, as used in this work, denotes a classical approximation strategy in which a non-quantum model is trained to reproduce selected task-relevant behaviors associated with a quantum-inspired reference model. These behaviors may include forecasting responses, output-distribution alignment, spectral-structure similarity, temporal-correlation preservation, and observable-level agreement. This does not imply quantum-state equivalence, genuine entanglement reproduction, or quantum computational advantage.

This distinction is essential for the present work. The proposed CeNN framework belongs to the third category. It does not simulate the full quantum state more efficiently. Instead, it changes the approximation target from state fidelity to task-relevant functional fidelity. This is the basis for studying CeNN as a scalable classical emulator of selected QELM behaviors.

D. QUANTUM MACHINE LEARNING PARADIGMS FOR TSF

QML combines quantum information processing with learning algorithms. In hybrid QML models, classical data are encoded into quantum states, processed by quantum circuits or quantum dynamical systems, and measured to generate classical outputs for classification, regression, or forecasting [5]–[7].

For TSF, QML-related approaches can be grouped into three categories:

- **Physical quantum models:** models intended to run directly on quantum hardware. These remain constrained by NISQ noise, limited qubit count, restricted connectivity, and circuit-depth limitations.

- **Hybrid quantum-classical models:** models in which quantum circuits act as feature maps, trainable modules, kernels, or reservoirs inside a classical optimization pipeline.
- **Quantum-inspired functional emulators:** classical models that do not perform quantum computation but approximate selected task-relevant behaviors of quantum or quantum-inspired reference models. The CeNN framework evaluated in this paper belongs to this category.

E. MAIN QML ALGORITHMIC FAMILIES IN TSF

The QML-TSF literature contains several algorithmic families.

Quantum Support Vector Machines (QSVMs) use quantum feature maps or quantum kernels to evaluate similarities in high-dimensional Hilbert spaces [8], [21]. They are typically attractive for small or medium-sized datasets with non-linear structure, but kernel estimation and hardware noise limit scalability.

Quantum Neural Networks (QNNs) and Variational Quantum Circuits (VQCs) use parameterized quantum circuits trained through classical optimizers [14], [22]. They are flexible but can suffer from barren plateaus, circuit-depth limitations, and high state-vector simulation cost.

Quantum Extreme Learning Machines (QELMs) and Emulated QELMs (EQELMs) use fixed or randomly initialized quantum-inspired feature maps with trainable classical readouts [23], [24]. Their appeal lies in fast readout training and reservoir-like feature generation, but performance may depend strongly on the quality of the fixed feature map.

Quantum Reservoir Computing (QRC) uses quantum dynamical systems as reservoirs whose internal evolution transforms temporal inputs into high-dimensional features [18], [19], [25]. QRC is well suited to dynamical systems but remains sensitive to reservoir configuration, measurement strategy, and hardware constraints.

Quantum-inspired graph, tensor, and operator approaches adapt quantum-inspired representations, tensor networks, or Schrödinger-inspired formulations to structured temporal data, including traffic and multivariate forecasting problems [26], [27].

This taxonomy provides the basis for the systematic review and for the later positioning of CeNN as a classical functional-emulation approach rather than a physical QML model.

III. SYSTEMATIC REVIEW METHODOLOGY

This systematic review was conducted and reported according to the PRISMA 2020 guidelines [17]. Its objective is to identify, classify, and critically synthesize studies applying Quantum Machine Learning (QML) or quantum-inspired methods to Time Series Forecasting (TSF). Particular attention is given to scalability, execution modality, baseline fairness, evaluation practice, and reproducibility.

TABLE 2. Representative search strings and database coverage.

Database	Representative Boolean query
IEEE Xplore	`(("quantum machine learning" OR QML) AND ("time series" OR forecasting) AND (QSVM OR QNN OR QELM OR QRC OR VQC))`
Scopus / WoS	`TITLE-ABS-KEY(("quantum machine learning") AND ("time series" OR "forecasting") AND (QSVM OR QNN OR QELM OR QRC OR VQC)) AND PUBYEAR > 2013`
arXiv	`all:("quantum machine learning" AND "time series") OR all:("quantum" AND forecasting)`
ScienceDirect	`("quantum machine learning" AND "time series" AND (VQC OR QNN OR QRC OR QELM))`

The review also serves a methodological role in this paper: it provides the evidence base used to motivate the bounded CeNN-based functional emulation framework developed in later sections. The review protocol specified the information sources, search terms, eligibility criteria, extraction variables, and risk-of-bias domains before the final screening stage. The protocol was not registered in a public registry; this limitation is acknowledged in Section VIII. The final search update and corpus freeze were performed on December 22, 2025.

A. SEARCH STRATEGY AND INFORMATION SOURCES

To cover both peer-reviewed publications and relevant preprints in the rapidly evolving QML field, seven electronic sources were queried: *IEEE Xplore*, *Scopus*, *Web of Science*, *SpringerLink*, *ScienceDirect*, *ACM Digital Library*, and *arXiv*. The inclusion of *arXiv* is justified by the fact that QML studies often circulate as preprints before formal journal or conference publication. Database searching was complemented by backward and forward snowballing using *Google Scholar*.

Search queries were adapted to the syntax of each database while preserving a common Boolean structure around three concepts: (i) quantum or quantum-inspired learning, (ii) time-series forecasting, and (iii) representative QML model families. Table 2 reports representative search strings.

B. SELECTION PROCESS AND ELIGIBILITY CRITERIA

The selection process followed four stages: identification, title-and-abstract screening, full-text eligibility assessment, and final inclusion. Two reviewers (G.N.D. and E.M.K.) independently screened the records. Disagreements were resolved by discussion, and unresolved cases were arbitrated by a senior reviewer (K.K.).

The inclusion criteria were:

- **IC1:** The study addresses TSF or a closely related temporal prediction task.

- **IC2:** The study uses a quantum, hybrid quantum-classical, or quantum-inspired machine learning method.
- **IC3:** The study reports quantitative forecasting metrics, such as RMSE, MAE, MAPE, sMAPE, or related error measures.
- **IC4:** The study includes at least one classical or non-quantum baseline.
- **IC5:** The study provides sufficient methodological detail to identify the model family, encoding strategy, execution modality, or computational backend.
- **IC6:** The study was published or made publicly available between January 2014 and December 2025.

The exclusion criteria were:

- **EC1:** The study does not involve TSF or temporal prediction.
- **EC2:** The study is purely theoretical or conceptual and does not include a forecasting experiment.
- **EC3:** The study does not report quantitative forecasting metrics.
- **EC4:** The QML or quantum-inspired model is not sufficiently identifiable.
- **EC5:** The study is a redundant preprint superseded by a later peer-reviewed or substantially updated version.
- **EC6:** The study provides insufficient information to assess model design, dataset use, or experimental comparison.

Figure 1 summarizes the selection process. From 1,248 initial records, 936 duplicates were removed, leaving 312 unique records. Title-and-abstract screening excluded 234 records, and 78 full-text reports were assessed for eligibility. Of these, 33 were excluded: 12 due to insufficient architectural transparency, 10 due to weak or missing classical baselines, 8 due to lack of quantitative TSF metrics, and 3 due to redundancy with later versions. The final corpus consisted of 45 studies included in the qualitative and quantitative synthesis.

C. DATA EXTRACTION AND CODING SCHEME

For each included study, we extracted bibliographic, methodological, experimental, and computational information. The extraction form included:

- **Bibliographic information:** authors, publication year, venue, and publication type.
- **Forecasting task:** application domain, dataset, forecasting horizon, univariate or multivariate setting, and temporal split strategy.
- **Model family:** QSVM, QNN/VQC, QLSTM/QGRU, QELM/EQELM, QRC, quantum kernel method, quantum-inspired graph model, or other hybrid architecture.
- **Encoding and architecture:** data-encoding strategy, number of qubits when applicable, circuit depth, ansatz type, reservoir configuration, or emulator design.
- **Execution modality:** classical simulation, quantum-inspired functional emulation, or physical quantum hardware execution.
- **Baselines and metrics:** classical baselines, hyperparameter-tuning protocol, reported metrics, statistical testing, and confidence intervals when available.
- **Reproducibility information:** code availability, dataset access, random seeds, preprocessing scripts, software framework, hardware backend, and configuration files.

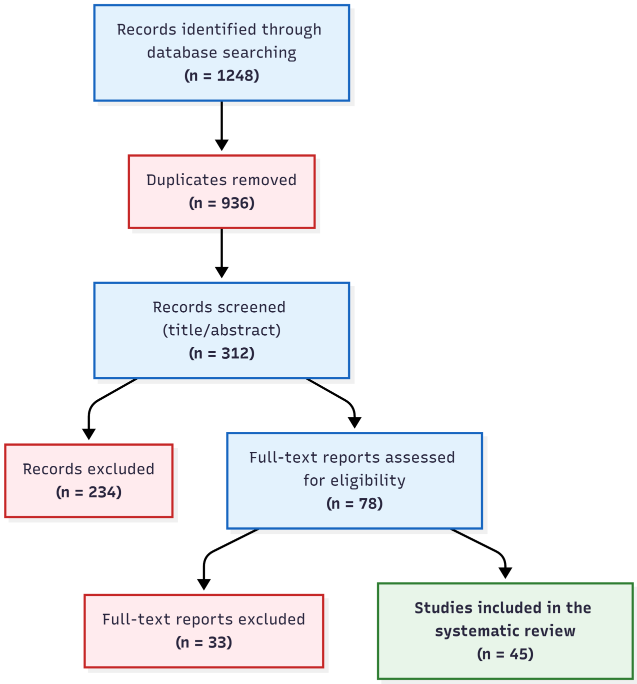

FIGURE 1. PRISMA 2020 flow diagram of the study selection process, from initial identification ($n = 1248$) to final inclusion ($n = 45$). Inter-rater agreement: Cohen's $\kappa = 0.85$.

This extraction scheme was designed to support both the systematic review and the later proof-of-concept study. In particular, the variables related to execution modality, scalability, and reproducibility were used to identify whether current QML-TSF approaches can be extended beyond small-scale simulation settings.

D. RISK OF BIAS AND QUALITY ASSESSMENT

To assess the methodological quality of the included studies, we developed a domain-specific risk-of-bias rubric for QML-TSF research. The rubric covers four domains: architectural transparency, baseline fairness, platform and noise characterization, and reproducibility. Two reviewers independently applied the rubric, and disagreements were resolved by consensus. Inter-rater reliability was substantial, with Cohen's $\kappa = 0.85$ and 95% CI [0.79, 0.91].

TABLE 3. Risk-of-bias assessment across the 45 included studies.

Quality domain	Low risk	Some concerns	High risk
D1: Architectural transparency	88% (40)	10% (4)	2% (1)
D2: Baseline rigor	76% (34)	20% (9)	4% (2)
D3: Platform and noise clarity	65% (29)	25% (11)	10% (5)
D4: Reproducibility	42% (19)	35% (16)	23% (10)

Numbers in parentheses indicate the number of studies in each category. Inter-rater agreement: Cohen's $\kappa = 0.85$.

1) Assessment Domains

D1: Architectural transparency. A study was rated as low risk if it provided a sufficiently complete description of the model architecture, including circuit diagram or gate sequence when applicable, data encoding, qubit count, circuit depth, ansatz or reservoir structure, parameter initialization, and optimizer. Studies with missing but non-critical elements were rated as some concerns, whereas vague or incomplete descriptions were rated as high risk.

D2: Baseline rigor and comparative fairness. A study was rated as low risk if it compared the proposed method against competitive classical baselines, such as LSTM, Transformer-type models, N-BEATS, N-HiTS, XGBoost, or relevant statistical baselines, with evidence of fair experimental setup or hyperparameter tuning. Incomplete reporting led to some concerns, while missing or clearly insufficient baselines led to high risk.

D3: Platform specification and noise characterization. A study was rated as low risk if it clearly reported whether results were obtained through simulation, emulation, or physical hardware execution. For simulation or hardware experiments, relevant software frameworks, backends, noise assumptions, shot counts, or hardware parameters were assessed when applicable.

D4: Data, code, and experimental reproducibility. A study was rated as low risk if the dataset, preprocessing protocol, train/validation/test split, random seeds, code repository, and configuration details were publicly available or sufficiently described. Incomplete availability led to some concerns, whereas missing or insufficient experimental details led to high risk.

2) Risk-of-Bias Results

Table 3 summarizes the risk-of-bias assessment across the 45 included studies. Reproducibility is the weakest domain, with only 42% of studies rated as low risk. This finding directly motivates the detailed implementation, configuration, and code-availability information provided for the CeNN proof of concept in Section IX.

E. SYNTHESIS STRATEGY

The synthesis combined qualitative taxonomy construction with quantitative performance-ratio analysis. Qualitatively, studies were grouped by model family, execution modality, encoding strategy, and application domain. This enabled the identification of recurring patterns, including small simulated systems, heterogeneous baselines, limited reporting of noise assumptions, and incomplete reproducibility information.

Quantitatively, when sufficient information was available, reported forecasting errors were normalized relative to the strongest classical baseline used in the same study:

$$r_m = \frac{E_{\mathrm{QML},m}}{E_{\mathrm{baseline},m}}, \tag{4}$$

where $E_{\mathrm{QML},m}$ is the reported error of the QML or quantum-inspired model under metric m, and $E_{\mathrm{baseline},m}$ is the corresponding error of the strongest classical baseline reported in the same experimental setting. A ratio $r_m < 1$ indicates that the QML or quantum-inspired model outperformed the baseline for that metric.

Because the included studies differ in datasets, horizons, metrics, and reporting practices, pooled estimates are interpreted as indicative trends rather than definitive effect sizes. Where applicable, geometric means and confidence intervals were used to summarize relative performance ratios. Studies with high risk in baseline fairness or reproducibility were analyzed separately in sensitivity checks.

F. FROM SYSTEMATIC REVIEW FINDINGS TO CENN FRAMEWORK DESIGN

A key purpose of the systematic review is to identify the methodological gaps that motivate the proposed CeNN-based functional emulator. Table 4 summarizes the link between review findings and framework design choices. This mapping does not imply that CeNN is the only possible response to the identified gaps; it explains why CeNN is investigated here as one bounded and testable functional-emulation pathway.

G. SENSITIVITY ANALYSIS

To assess whether the synthesis was disproportionately influenced by lower-quality studies, sensitivity checks were performed by excluding studies rated as high risk in either baseline rigor (D2) or reproducibility (D4). The aggregated relative performance ratios remained directionally stable, with a mean shift below 0.02 and no statistically significant change at the 5% level. These results suggest that the broad trends reported in the review are not solely driven by studies with weak baselines or poor reproducibility. However, given the heterogeneity of datasets, metrics, and reporting practices, these results are interpreted cautiously and are not treated as conclusive evidence of general QML superiority.

H. PRISMA 2020 COMPLIANCE STATEMENT

This review follows the PRISMA 2020 reporting structure to the extent applicable to a methodological review of QML-TSF. A completed PRISMA checklist mapping

TABLE 4. From systematic review findings to CeNN framework design choices.

Review finding	Limitation identified	CeNN design response
Many QML-TSF studies rely on small state-vector simulations.	State-vector methods face exponential memory growth as the number of qubits increases.	Use a locally connected CeNN representation whose architectural parameter growth is controlled by the number of cells and bounded neighborhood size.
NISQ hardware remains affected by noise, decoherence, and limited connectivity.	Simulation results may not transfer directly to physical quantum hardware.	Position CeNN as a classical functional emulator, not as a substitute for physical quantum execution.
Benchmarking practices are heterogeneous across studies.	Claims of QML superiority are difficult to compare across datasets, horizons, and baselines.	Evaluate the proof of concept against naive baselines, MLP, LSTM, Transformer, and a QELM reference model under a common chronological split.
Reproducibility reporting is incomplete in many studies.	Missing code, seeds, preprocessing details, and configuration information limit independent verification.	Report implementation details, random seeds, configuration settings, exported outputs, and code availability for the CeNN experiments.
Interpretability and constraint handling remain underdeveloped.	Black-box quantum or hybrid models are difficult to audit in forecasting applications.	Include differentiable regularization and emulation losses to study the trade-off between predictive accuracy and functional alignment.

each item to its corresponding manuscript location is provided in Appendix D. The review protocol was not registered in a public registry, which is acknowledged as a methodological limitation. All search, screening, extraction, and quality-assessment decisions are reported to support transparency and reproducibility.

IV. QUANTUM ALGORITHMS FOR TIME SERIES FORECASTING: A TAXONOMY

This section synthesizes the reviewed Quantum Machine Learning (QML) and quantum-inspired approaches for Time Series Forecasting (TSF). The purpose is not to claim a general quantum advantage over classical forecasting models, but to identify how different quantum, hybrid quantum-classical, and quantum-inspired methods have been used, evaluated, and constrained in the TSF literature.

Consistent with the distinction introduced in Section II-C, the reviewed methods are analyzed according to four criteria: learning mechanism, encoding strategy, execution modality, and practical forecasting limitations. The taxonomy provides the basis for identifying the scalability, benchmarking, and reproducibility gaps that motivate the CeNN-based functional emulator introduced later in the paper.

The reviewed literature does not support a single uniform conclusion about QML superiority. Instead, it reveals a heterogeneous landscape in which quantum and quantum-inspired models may be useful under specific data regimes, model capacities, and execution assumptions. Some approaches rely on quantum feature spaces or parameterized quantum circuits, whereas others use reservoir dynamics, fixed quantum-inspired mappings, or classical functional emulation.

Because the reviewed studies differ in datasets, horizons, metrics, baselines, and execution platforms, the synthesis below is interpreted as a taxonomy of methodological patterns rather than as definitive evidence of general QML superiority.

TABLE 5. Qualitative patterns observed across QML and quantum-inspired families for TSF.

Family	Typical use in TSF	Main strengths	Main limitations
QSVM	Small or medium-sized non-linear forecasting tasks	Expressive kernel representation; clear mathematical basis	Kernel estimation cost; scalability with sample size and feature dimension
QNN/VQC	Hybrid forecasting with trainable quantum blocks	Flexible feature maps; hybrid optimization compatibility	Barren plateaus; NISQ noise; simulation overhead
QELM/EQELM	Fast forecasting with fixed or emulated hidden mappings	Closed-form readout; reduced training cost	Sensitivity to feature-map design and random projections
QRC	Dynamical and chaotic time-series modeling	Reservoir memory; few trainable parameters	Sensitivity to reservoir configuration and measurement protocol
Quantum-inspired emulators	Classical approximation of selected quantum-inspired behaviors	Scalable classical execution; improved reproducibility	Functional approximation only; no physical quantum equivalence

A. QUANTUM SUPPORT VECTOR MACHINES

Quantum Support Vector Machines (QSVMs) extend the classical SVM framework by replacing or augmenting the classical kernel with a quantum kernel. Given an input vector $x \in \mathbb{R}^d$, a quantum feature map $U_\phi(x)$ encodes the input into an n-qubit state:

$$|\phi(x)\rangle = U_\phi(x)|0\rangle^{\otimes n}. \tag{5}$$

The associated quantum kernel is commonly defined as

$$k_{\mathrm{Q}}(x, x') = |\langle \phi(x) \mid \phi(x')\rangle|^2 . \tag{6}$$

For TSF, x usually represents a lagged temporal window rather than an independent observation. In a regression setting, the predictive function can be written as

$$f(x) = \sum_i (\alpha_i - \alpha_i^*)\, k_{\mathrm{Q}}(x_i, x) + b, \tag{7}$$

where k_{Q} is the quantum kernel, α_i and α_i^* are support-vector coefficients, and b is the bias term.

The reviewed QSVM studies mainly use amplitude encoding, angle encoding, or structured feature maps such as Pauli or `ZZFeatureMap`-type encodings. QSVMs have been applied to financial forecasting, energy demand prediction, and hybrid kernel-recurrent models [10], [28]–[30]. Their strength lies in their clear kernel-based formulation. Their main limitation is scalability: kernel matrix construction becomes expensive as the number of samples grows, and real-device execution introduces shot noise, gate errors, and hardware-dependent variability.

B. QUANTUM NEURAL NETWORKS AND VARIATIONAL QUANTUM CIRCUITS

Quantum Neural Networks (QNNs), commonly implemented as Variational Quantum Circuits (VQCs), are among the most frequently studied QML models in TSF. They combine quantum circuits with classical optimization. A typical VQC-based forecasting model includes data encoding, a parameterized quantum ansatz, measurement, and classical optimization.

The input encoding stage maps a temporal vector x into a quantum state $|\psi(x)\rangle = U_\phi(x)|0\rangle$. A trainable unitary U_θ then applies parameterized rotations and, when applicable, entangling gates. The model output is obtained from measured expectation values such as $\langle \hat{O} \rangle$, which are used directly as predictions or as features for a classical readout.

Several VQC variants have been explored for temporal data. Quantum LSTM (QLSTM) models insert VQC blocks into LSTM-like gating mechanisms [11]. Quantum GRU models follow a similar idea with fewer gating parameters. Quantum CNNs and data re-uploading circuits attempt to increase expressivity under limited qubit counts [22].

QNN/VQC models are flexible and expressive, but their practical use in TSF remains constrained. First, optimization may suffer from barren plateaus as circuit depth or qubit count increases [14]. Second, state-vector simulation becomes expensive beyond small qubit regimes. Third, physical NISQ execution is affected by decoherence, finite-shot noise, gate errors, and limited connectivity. For this reason, most QNN/VQC forecasting results should be interpreted as controlled experimental evidence rather than as mature large-scale deployment results.

C. QUANTUM EXTREME LEARNING MACHINES AND EMULATED VARIANTS

Quantum Extreme Learning Machines (QELMs) adapt the Extreme Learning Machine paradigm to quantum or quantum-inspired feature mappings. In this family, the hidden transformation is fixed, randomly initialized, or generated by a quantum-inspired mapping, while only the output readout is learned. This reduces training cost compared with fully trainable variational circuits.

A simplified QELM formulation maps an input x into hidden features $h_i(x)$ using a fixed transformation:

$$h_i(x) = \langle \psi_i | U_\phi(x) | 0 \rangle, \tag{8}$$

where $U_\phi(x)$ is an input-dependent transformation and $|\psi_i\rangle$ denotes a fixed reference state or projection direction. The prediction is obtained through a linear readout:

$$y(x) = \sum_{i=1}^{N_h} \beta_i h_i(x), \qquad \boldsymbol{\beta} = H^{+}\mathbf{Y}, \tag{9}$$

where N_h is the number of hidden features, H^+ is the Moore–Penrose pseudoinverse of the hidden-feature matrix, and $\mathbf{Y}$ is the target vector.

QELM and emulated QELM variants are attractive because they can train quickly and avoid full iterative optimization of all hidden parameters [23], [24]. Recent long-term forecasting frameworks such as QuLTSF illustrate the growing interest in quantum-inspired mappings for TSF [24]. Density-matrix emulation approaches also suggest that functional emulation can serve as a methodological bridge when exact simulation or direct hardware execution is impractical [13].

The main limitation is that QELM performance depends strongly on the quality of the fixed feature map, the random projection mechanism, the number of hidden features, and the regularization of the readout. Because the hidden mapping is often fixed, QELMs may be less adaptive than fully trainable deep models. This sensitivity motivates the use of QELM in this paper as a measurable reference model rather than as an assumed superior predictor.

D. QUANTUM RESERVOIR COMPUTING

Quantum Reservoir Computing (QRC) extends reservoir computing to quantum dynamical systems. As in Echo State Networks or Liquid State Machines, the internal reservoir is fixed or only weakly trained, while a classical readout is optimized. This makes QRC relevant for TSF because reservoir models are designed to transform temporal inputs into state trajectories with fading memory [18], [19], [25].

A typical QRC state may be represented by a density matrix ρ_t evolving as

$$\rho_t = U(x_t)\rho_{t-1}U^\dagger(x_t), \qquad U(x_t) = e^{-iH(x_t)\Delta t}, \tag{10}$$

where $H(x_t)$ is an input-modulated Hamiltonian and Δt is the evolution time step. The output is obtained from expectation values:

$$y_t = \mathrm{Tr}(O\rho_t) = \langle O \rangle_t. \tag{11}$$

QRC has been explored for chaotic systems, volatility modeling, and temporal prediction under complex dynamics [31]–[33]. Its strengths are dynamical memory and relatively simple readout training. Its limitations include sensitivity to reservoir design, input injection, measurement protocol, and hardware or simulation constraints. QRC is therefore promising for dynamical TSF, but its reported performance remains strongly configuration-dependent.

Quantum Reinforcement Learning (QRL) is sometimes discussed in relation to adaptive decision-making and control. However, it is not a central TSF model family in the reviewed corpus. It is therefore treated only as a peripheral future direction for adaptive forecasting systems in which prediction and decision-making are jointly optimized.

E. HYBRID CLASSICAL–QUANTUM FORECASTING ARCHITECTURES

Hybrid classical–quantum architectures are among the most practical forms of QML experimentation in TSF. In these models, classical components handle preprocessing, feature extraction, dimensionality reduction, or output decoding, while quantum circuits or quantum-inspired modules act as specialized feature transformations.

A common hybrid pipeline contains three stages. First, a classical encoder, such as an MLP, CNN, recurrent layer, or transformer-like encoder, compresses the temporal input into a latent vector. Second, a quantum or quantum-inspired module, such as a VQC, quantum kernel, QELM, QRC, or emulator, transforms this latent representation. Third, a classical decoder maps the measured or emulated features to the forecasting target.

Quantum Transfer Learning (QTL) has also been explored as a strategy for adapting parameters or representations from a source task to a target task [34]. In TSF, QTL may be useful when target-domain data are scarce. However, non-stationarity and concept drift can produce negative transfer when the temporal dynamics of the source and target domains differ.

Hybrid models should therefore be viewed as experimental bridges rather than established industrial forecasting standards. Their advantages depend on fair comparison with strong classical baselines, careful hyperparameter selection, and transparent reporting of simulation or hardware assumptions.

F. APPLICATION DOMAINS OF QML-TSF MODELS

The reviewed literature applies QML and quantum-inspired models to several forecasting domains. The term application domain is used here instead of industrial deployment, because most studies remain experimental or proof-of-concept in nature.

Finance and market forecasting. Financial time series are noisy, non-stationary, and characterized by regime shifts. Several studies have explored quantum kernels, QNNs, QRC, or hybrid models for stock-price forecasting, realized-volatility prediction, and market trend analysis [32], [35]–[37]. Reported results remain sensitive to data splits, transaction-cost assumptions, baselines, and robustness to market-regime changes.

Energy systems and smart grids. Energy load forecasting, renewable generation prediction, and demand-response modeling are common QML-TSF application areas [10], [11], [29], [38]–[41]. These tasks are relevant because they combine seasonality, weather dependence, non-linearity, and operational constraints. However, many studies still rely on limited datasets or small simulated quantum models.

Traffic, mobility, and spatio-temporal forecasting. Traffic forecasting has been explored through quantum-inspired graph models, quantum neural networks, and hybrid architectures [22], [26], [42]. These problems are naturally spatio-temporal, but current comparisons with modern graph neural networks and transformer-based traffic models remain limited.

Healthcare and physiological signals. Some studies examine QML-inspired models for physiological or healthcare time series [43]. Such applications require particular caution because interpretability, reliability, and validation under realistic conditions are essential. At present, most results should be treated as exploratory rather than clinically deployable.

G. META-SYNTHESIS OF QML FOR TSF

The review shows that QML-TSF is not a homogeneous research direction. Rather than replacing classical forecasting models, QML and quantum-inspired approaches currently function as specialized modeling strategies whose usefulness depends on the forecasting task, encoding choice, execution modality, and baseline strength.

A recurring pattern is the trade-off between representational richness and scalability. QSVMs provide a clear kernel-based formulation but face kernel estimation and Gram-matrix scaling costs. QNNs and VQCs provide flexible trainable quantum modules but are constrained by barren plateaus, NISQ noise, and simulation overhead. QELM/EQELM models reduce training cost through fixed hidden transformations and analytical readouts, but are sensitive to feature-map design. QRC is naturally suited to temporal dynamics, but depends strongly on reservoir configuration and measurement strategy. Quantum-inspired emulators improve classical scalability by abandoning state-level fidelity, but must

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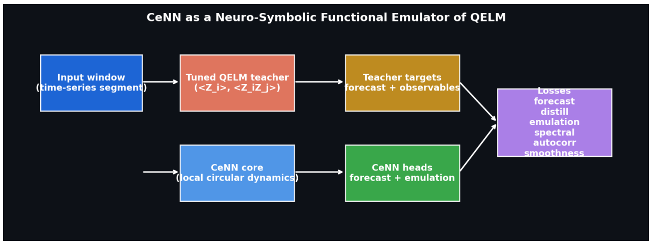

FIGURE 2. Overview of the proposed CeNN-based functional-emulation framework. A QELM reference model provides prediction targets and observable-level features, while the CeNN produces both forecasting outputs and emulation outputs. The framework is evaluated as a classical functional emulator, not as a quantum-state simulator.

specify exactly which functional behaviors are approximated and how those behaviors are validated.

The reviewed corpus also indicates that many promising TSF results are obtained in hybrid or emulated settings rather than through large-scale physical quantum execution. This does not demonstrate physical quantum advantage. Instead, it suggests that current QML-TSF research often uses quantum-inspired representations, simulated circuits, or emulated dynamics as experimental tools for representation learning.

This observation directly motivates the proposed CeNN framework. The framework is not intended to replace all QML families, nor does it claim physical equivalence with quantum models. Its purpose is to evaluate whether a locally connected neuro-symbolic dynamical system can approximate selected functional behaviors associated with QELM-style forecasting while remaining computationally tractable on classical hardware.

V. NEURO-SYMBOLIC CENN FRAMEWORK FOR QUANTUM-INSPIRED FUNCTIONAL EMULATION

The systematic review conducted in the preceding sections identifies a recurring methodological gap in QML-based time series forecasting: many quantum and hybrid models are expressive, but their practical evaluation is often constrained by small-scale simulation, NISQ noise, heterogeneous baselines, and incomplete reproducibility [12], [14], [15]. In response to this gap, this section introduces a bounded neuro-symbolic Cellular Neural Network (CeNN) framework for quantum-inspired functional emulation.

The proposed framework is classical. It does not simulate a full quantum state, reproduce genuine entanglement, or claim physical quantum computational advantage. Instead, it evaluates whether a locally connected dynamical architecture, guided by forecasting, emulation, spectral, temporal, and regularization objectives, can approximate selected task-relevant behaviors of a Quantum Extreme Learning Machine (QELM) reference model [23], [24].

A. PROBLEM SETTING AND SCOPE

Let $\mathbf{X}_t = [x_{t-L+1}, \ldots, x_t]$ denote a temporal input window of length L, and let $\mathbf{y}_{t+1:t+h}$ denote the forecasting target over horizon h. The objective is to learn a forecasting map

$$F_\theta : \mathbf{X}_t \mapsto \hat{\mathbf{y}}_{t+1:t+h}, \tag{12}$$

where θ denotes the trainable parameters of the CeNN-based emulator.

The guiding question is whether a classical locally connected dynamical model can approximate selected forecasting-relevant behaviors of a QELM reference model while avoiding the exponential memory growth associated with full state-vector simulation. The term functional emulation is used deliberately: the framework targets output-level, observable-level, and task-level behaviors, not quantum-state fidelity.

The CeNN emulator is evaluated through three main dimensions:

1) forecasting utility on selected time-series datasets;
2) functional alignment with QELM predictions and enriched QELM observables;
3) architectural scalability under local-connectivity assumptions.

B. FROM STATE FIDELITY TO FUNCTIONAL FIDELITY

A full state-vector simulation of an n-qubit quantum system requires storing 2^n complex amplitudes. Density-matrix representations are even more expensive, since they may require $2^n \times 2^n$ complex entries for mixed states. This exponential scaling limits direct simulation as the number of qubits or effective latent dimensions increases [14], [15].

The CeNN framework avoids this representation by changing the approximation target. Instead of attempting to reproduce the complete quantum state, it uses a classical state vector of N real-valued cells:

$$\mathbf{s}(t) = [s_1(t), s_2(t), \ldots, s_N(t)]^\top \in \mathbb{R}^N. \tag{13}$$

Each cell interacts only with a bounded local neighborhood $\mathcal{N}(i)$ determined by a fixed radius. Under this bounded-neighborhood assumption, the number of interactions per cell remains constant as N grows. This locality is the basis for the linear architectural scaling discussed in Section V-G.

Importantly, CeNN cells are not qubits. They are classical dynamical units. Their role is to provide a scalable substrate for approximating selected functional behaviors useful for forecasting.

C. QELM REFERENCE MODEL AND OBSERVABLE TARGETS

The reference model used in the proof-of-concept study is a QELM-style teacher. QELM models adapt the

Extreme Learning Machine principle to quantum or quantum-inspired feature mappings, typically using fixed hidden transformations and a trainable classical readout [23], [24]. In this paper, the role of the QELM is not to provide a ground-truth quantum state, but to provide measurable reference behaviors against which functional emulation can be assessed.

For an input window $\mathbf{X}_t$, the QELM teacher produces two types of targets:

1) a multi-step prediction $\hat{\mathbf{y}}^{\mathrm{QELM}}_{t+1:t+h}$;
2) an enriched observable vector $\mathbf{o}^{\mathrm{QELM}}_t$.

The enriched observable vector contains first-order and pairwise Z-observable terms:

$$\mathbf{o}^{\mathrm{QELM}}_t = [\langle Z_i \rangle, \langle Z_i Z_j \rangle], \tag{14}$$

where $\langle Z_i \rangle$ denotes the expectation of a single-qubit observable and $\langle Z_i Z_j \rangle$ denotes a pairwise observable. These observables provide measurable emulation targets for the CeNN emulation head.

If the QELM teacher performs poorly on validation relative to a naive baseline, its influence is down-weighted in the training objective. This teacher-quality control prevents the CeNN from being forced to imitate an uninformative reference model.

D. CENN DYNAMICAL LAYER

At the core of the framework, the CeNN is modeled as a locally connected nonlinear dynamical system. Following the classical CeNN formulation of Chua and Yang [16], [44], the state of cell i evolves as

$$\dot{s}_i(t) = -s_i(t) + \sum_{j\in\mathcal{N}(i)} A_{ij} z_j(t) + \sum_{j\in\mathcal{N}(i)} B_{ij} u_j(t) + I_i, \tag{15}$$

where $s_i(t)$ is the internal state of cell i, $z_j(t)$ is the output of neighboring cell j, $u_j(t)$ is an external input, A_{ij} and B_{ij} are feedback and control templates, and I_i is a bias term.

The cell output is obtained through a bounded nonlinear activation:

$$z_i(t) = \sigma(s_i(t)). \tag{16}$$

For forecasting, the final hidden CeNN state is passed to a forecasting head:

$$\hat{\mathbf{y}}_{t+1:t+h} = g_W(\mathbf{s}(t)), \tag{17}$$

where g_W is a trainable classical readout.

The implemented architecture also includes an emulation head:

$$\hat{\mathbf{o}}^{\mathrm{CeNN}}_t = g_E(\mathbf{s}(t)), \tag{18}$$

where $\hat{\mathbf{o}}^{\mathrm{CeNN}}_t$ approximates the enriched QELM observable vector in Eq. (14). This separation allows the framework to distinguish forecasting performance from functional emulation fidelity.

E. NEURO-SYMBOLIC AND EMULATION OBJECTIVES

The training objective combines direct forecasting with optional functional emulation and regularization terms. The neuro-symbolic component is used as a structured regularization mechanism, not as a claim of quantum behavior [45], [46]. The forecasting loss is defined as

$$\mathcal{L}_{\mathrm{data}} = \frac{1}{M}\sum_{m=1}^{M} \|\hat{\mathbf{y}}_m - \mathbf{y}_m\|_2^2 . \tag{19}$$

The distillation loss aligns CeNN predictions with QELM predictions:

$$\mathcal{L}_{\mathrm{distil}} = \frac{1}{M}\sum_{m=1}^{M} \left\|\hat{\mathbf{y}}^{\mathrm{CeNN}}_m - \hat{\mathbf{y}}^{\mathrm{QELM}}_m\right\|_2^2 . \tag{20}$$

The observable-emulation loss aligns the CeNN emulation head with the QELM observable vector:

$$\mathcal{L}_{\mathrm{emul}} = \frac{1}{M}\sum_{m=1}^{M} \left\|\hat{\mathbf{o}}^{\mathrm{CeNN}}_m - \mathbf{o}^{\mathrm{QELM}}_m\right\|_2^2 . \tag{21}$$

Spectral alignment is used to compare frequency-domain structure between reference and emulated trajectories. Let $\Phi_{\mathrm{ref}}(\omega)$ and $\Phi_{\mathrm{CeNN}}(\omega)$ denote spectral representations of a reference trajectory and the CeNN trajectory. A spectral similarity index is defined as

$$\mathrm{SSI} = \frac{\int \Phi_{\mathrm{ref}}(\omega)\Phi_{\mathrm{CeNN}}(\omega)\, d\omega}{\|\Phi_{\mathrm{ref}}\|_2\, \|\Phi_{\mathrm{CeNN}}\|_2}, \tag{22}$$

with loss

$$\mathcal{L}_{\mathrm{spectral}} = 1 - \mathrm{SSI}. \tag{23}$$

Temporal-correlation preservation is assessed through autocorrelation profiles. For a bounded temporal feature $f(t)$, the lag-τ autocorrelation is

$$C_f(\tau) = \mathbb{E}[f(t)f(t+\tau)]. \tag{24}$$

The correlation-preservation loss is

$$\mathcal{L}_{\mathrm{corr}} = \sum_{\tau\in\mathcal{T}} \left|C_{\mathrm{ref}}(\tau) - C_{\mathrm{CeNN}}(\tau)\right|^2 , \tag{25}$$

where $\mathcal{T}$ is a selected set of lags.

A smoothness regularization term is used as a lightweight symbolic constraint:

$$\mathcal{L}_{\mathrm{smooth}} = \sum_t \left(\hat{y}_t - \hat{y}_{t-1}\right)^2 . \tag{26}$$

Other constraints, such as non-negativity or capacity limits, are not activated by default in the benchmark because the models are trained in standardized space. This avoids penalizing values that are negative only because of normalization rather than because of physical invalidity.

The total objective is

$$\begin{aligned}\mathcal{L}_{\mathrm{total}} =& \mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{distil}}\mathcal{L}_{\mathrm{distil}} + \lambda_{\mathrm{emul}}\mathcal{L}_{\mathrm{emul}} \\ &+ \lambda_{\mathrm{spectral}}\mathcal{L}_{\mathrm{spectral}} + \lambda_{\mathrm{corr}}\mathcal{L}_{\mathrm{corr}} + \lambda_{\mathrm{smooth}}\mathcal{L}_{\mathrm{smooth}}.\end{aligned} \tag{27}$$

Algorithm 1 Training procedure of the neuro-symbolic CeNN emulator

Require: Training data $\mathcal{D} = \{(\mathbf{X}_i, \mathbf{y}_i)\}_{i=1}^{M}$, QELM teacher outputs, loss weights λ
Ensure: Learned CeNN parameters θ

```
Fit or select the QELM teacher on the validation split
Generate QELM prediction targets and observable targets
Initialize CeNN parameters θ
for epoch = 1 to E do
    for mini-batch (X, y) in D do
        Compute CeNN forecast ŷ^CeNN
        Compute CeNN emulation output ô^CeNN
        Compute L_data
        Compute L_distil if the QELM teacher is informative
        Compute L_emul using QELM observables
        Compute spectral, correlation, and smoothness terms
        Compute L_total using Eq. (27)
        Update θ by gradient-based optimization
    end for
end for
```

The weights λ_{distil}, λ_{emul}, $\lambda_{\text{spectral}}$, λ_{corr}, and λ_{smooth} control the contribution of each term. In the experiments, two CeNN variants are evaluated: a balanced CeNN_NS model and a more emulation-oriented CeNN_Emulator. This distinction prevents forecasting accuracy and QELM emulation from being conflated.

F. ERROR-BOUND INTERPRETATION

The distributional alignment objective allows a bounded interpretation of output-level approximation error. Let P_{ref} be a reference output distribution and Q_θ the CeNN-induced distribution. For any bounded observable f satisfying $|f(x)| \leq B$, Pinsker's inequality gives

$$|\mathbb{E}_{P_{\text{ref}}}[f] - \mathbb{E}_{Q_\theta}[f]| \leq B\sqrt{2D_{\text{KL}}(P_{\text{ref}} \parallel Q_\theta)}. \tag{28}$$

Thus, if

$$D_{\text{KL}}(P_{\text{ref}} \parallel Q_\theta) \leq \epsilon, \tag{29}$$

then

$$|\mathbb{E}_{P_{\text{ref}}}[f] - \mathbb{E}_{Q_\theta}[f]| \leq B\sqrt{2\epsilon}. \tag{30}$$

For temporal correlations involving bounded functions, an analogous interpretation gives

$$|\mathbb{E}_{P_{\text{ref}}}[f(t)f(t+\tau)] - \mathbb{E}_{Q_\theta}[f(t)f(t+\tau)]| \leq B^2\sqrt{2\epsilon}. \tag{31}$$

These inequalities do not prove that the CeNN reproduces quantum dynamics. They only show that, if the reference and emulated output distributions are close in KL divergence, then bounded output-level observables and second-order temporal statistics are also close. This is the level of approximation claimed in this paper.

TABLE 6. Illustrative memory comparison between state-vector simulation and CeNN functional emulation.

n or N	State-vector simulation	CeNN representation
16	≈ 1 MB	KB-scale
32	≈ 64 GB	KB-scale
48	≈ 4 PB	KB-scale

G. FORMAL COMPLEXITY ANALYSIS

The scalability claim of the CeNN emulator relies on explicit local-connectivity assumptions. Let N be the number of CeNN cells, let r be the local connectivity radius, and let $k = 2r + 1$ denote the maximum number of neighbors per cell in the one-dimensional circular implementation used in the experiments. The radius r is fixed independently of N, so k remains constant as N grows.

Each cell stores one real-valued state and a bounded number of local template parameters. Therefore, the memory requirement is

$$M_{\text{CeNN}} = \mathcal{O}(N) + \mathcal{O}(Nk) = \mathcal{O}(N), \tag{32}$$

because k is constant.

For each update, each cell aggregates information from at most k neighbors. The per-cell update cost is $\mathcal{O}(k)$, and the total update cost is therefore

$$T_{\text{CeNN}} = \mathcal{O}(Nk) = \mathcal{O}(N). \tag{33}$$

By contrast, an n-qubit state-vector simulator requires

$$M_{\text{sim}} = \mathcal{O}(2^n) \tag{34}$$

complex amplitudes. In double precision, this corresponds approximately to

$$M_{\text{sim}} \approx 16 \times 2^n \text{ bytes}. \tag{35}$$

This comparison is not a claim that CeNN simulates the same quantum state more efficiently. It states that the CeNN changes the target from state-vector fidelity to task-relevant functional fidelity.

H. COMPLEXITY ILLUSTRATION

Table 6 illustrates the memory contrast between state-vector simulation and a locally connected CeNN representation under simplified assumptions.

The table is intended as a scaling illustration, not as proof of physical equivalence between the two representations.

I. POSITIONING RELATIVE TO OTHER QML FAMILIES

The choice of CeNN is motivated by the systematic review findings, but it is not presented as the only possible response to QML-TSF scalability limitations. Table 7 summarizes its positioning relative to selected model families.

TABLE 7. Positioning of the CeNN emulator relative to selected QML-TSF model families.

Family	Strength	Limitation	CeNN-related rationale
QSVM	Strong kernel formulation	Kernel estimation and scaling cost	CeNN avoids Gram-matrix construction and targets dynamical forecasting behavior.
QNN/VQC	Flexible trainable quantum circuits	Barren plateaus, simulation cost, and NISQ noise	CeNN avoids circuit-depth constraints by using classical local dynamics.
QELM/EQELM	Fast readout and reduced training cost	Sensitivity to random or fixed features	CeNN uses QELM as a measurable reference while adding trainable local dynamics.
QRC	Natural temporal memory	Reservoir tuning and hardware dependence	CeNN provides a classical reservoir-like substrate with explicit local connectivity.
CeNN emulator	Local update, symbolic regularization, and reproducibility	Functional approximation only; no quantum-state fidelity	Investigated as a bounded classical pathway for scalable functional emulation.

J. EXPERIMENTAL ROLE OF THE FRAMEWORK

The detailed experimental protocol and results are presented in Section VI. To avoid over-claiming, the experiments are treated as a bounded proof of concept rather than as definitive validation of generality, hardware transferability, or industrial readiness.

The final benchmark uses four datasets covering energy load forecasting, financial log-returns, weather temperature, and deterministic chaotic dynamics. Specifically, the benchmark includes Energy, AAPL log-returns, Jena Climate, and Mackey–Glass. Three datasets correspond to real-world time series, while Mackey–Glass is included as a controlled chaotic benchmark.

The protocol uses 20 000 chronological observations per dataset when available, a 70/15/15 chronological train-validation-test split, a sliding input window of 48 observations, a 24-step forecasting horizon, and five random seeds for trainable models. The evaluation reports forecasting accuracy, functional emulation fidelity, statistical comparison, component sensitivity, inference latency, training time, and reproducibility metadata.

The final results show domain-dependent behavior. CeNN_NS achieves competitive forecasting performance against strong neural baselines, but it does not uniformly outperform MLP or LSTM across all datasets. Functional emulation is strongest on the Mackey–Glass chaotic benchmark and weaker on energy and weather series. This empirical pattern supports the feasibility of CeNN-based functional emulation as a research direction, while also reinforcing the need for broader validation across datasets, horizons, and hardware-aware settings.

K. SYNTHESIS AND RESEARCH IMPLICATIONS

The proposed CeNN framework provides a classical, locally connected, neuro-symbolic substrate for studying quantum-inspired functional emulation in TSF. Its main contribution is not to reproduce quantum mechanics, but to offer a scalable and reproducibility-oriented approximation framework targeting selected forecasting-relevant behaviors: output structure, observable-level alignment, temporal dynamics, spectral patterns, and domain-aware regularization.

The full benchmark strengthens the empirical basis of this framework by evaluating it on four domains under a common protocol. The results indicate that CeNN-based functional emulation can be competitive, particularly as a forecasting model with local architectural scaling. However, they also show that CeNN does not uniformly dominate classical neural baselines and that QELM alignment remains domain-dependent. This is an important finding because it prevents over-generalization and clarifies the current scope of the contribution.

From a research perspective, the framework supports four implications. First, it provides a tractable way to study quantum-inspired functional behaviors without requiring full state-vector simulation or physical QPU access. Second, it offers a locally connected dynamical alternative to dense architectures, with memory and update costs that scale linearly under bounded-neighborhood assumptions. Third, it allows emulation and regularization terms to be incorporated into forecasting models, although the ablation results show that these components should not be assumed to improve raw accuracy in all domains. Fourth, it provides a reproducible experimental basis for comparing functional emulation against classical forecasting baselines under matched preprocessing, splitting, and evaluation protocols.

The limitations of the framework are equally important. The CeNN emulator does not reproduce genuine entanglement, phase-coherent quantum interference, or full quantum-state evolution. It does not demonstrate physical quantum advantage and should not be interpreted as a replacement for quantum hardware. Broader validation is required on additional datasets, multivariate benchmarks, longer horizons, noisy settings, and regime-shift scenarios.

Overall, the proposed framework should be interpreted as a bounded methodological contribution: it provides a scalable classical pathway for studying quantum-inspired functional emulation in TSF, while the empirical findings show both its promise and its limits.

VI. PROOF-OF-CONCEPT BENCHMARKING AND DISCUSSION

This section reports the proof-of-concept evaluation of the proposed CeNN-based functional-emulation framework. The objective is not to establish universal superiority over classical or quantum-inspired models, but to evaluate three specific questions: whether the CeNN provides competitive forecasting performance, whether it exhibits measurable functional alignment with a QELM reference model, and whether its local architecture supports near-linear scaling under bounded-neighborhood assumptions [16], [23], [24].

A. EXPERIMENTAL PROTOCOL AND REPRODUCIBILITY

The empirical evaluation was conducted on four time-series datasets covering energy load forecasting, financial log-return forecasting, weather temperature forecasting, and deterministic chaotic dynamics. The benchmark includes Energy, AAPL log-returns, Jena Climate, and Mackey–Glass. Energy, AAPL, and Jena correspond to real-world time series, while Mackey–Glass is included as a controlled chaotic benchmark [47].

Each dataset was processed chronologically. No random shuffling was applied. For each dataset, up to 20 000 chronological observations were used after cleaning and transformation. The temporal split was fixed at 70% for training, 15% for validation, and 15% for testing. Normalization parameters were fitted only on the training split and then applied to the validation and test splits, preventing lookahead bias. This chronological evaluation design follows standard precautions for time series forecasting [1], [48].

The forecasting task used a 48-step input window and a 24-step forecasting horizon. Trainable models were evaluated over five random seeds:

$$\mathcal{S} = \{42, 123, 2024, 7, 99\}. \tag{36}$$

The primary forecasting metric was MASE, because it enables comparison against a naive baseline and is more robust than percentage-based metrics when the target can be close to zero [49]. RMSE, MAE, and R^2 were used as supporting metrics. Functional emulation was evaluated using Pearson correlation, KL divergence, spectral distance, autocorrelation-profile distance, observable-feature MSE, and mean observable correlation.

The evaluated models were selected to provide a balanced comparison across naive, neural, quantum-inspired, and CeNN-based approaches. Persistence and moving average serve as simple forecasting baselines. MLP, LSTM, and Transformer represent standard neural baselines [3], [4]. The QELM teacher provides a measurable quantum-inspired reference model [23], [24]. CeNN_NS is the balanced model optimized for forecasting and partial emulation, while CeNN_Emulator is used to study functional alignment with the QELM reference.

TABLE 8. Proof-of-concept experimental protocol.

Item	Protocol
Evaluation type	Bounded proof-of-concept benchmarking
Datasets	Energy, AAPL log-returns, Jena Climate, Mackey–Glass
Observations	Up to 20 000 chronological observations per dataset
Temporal split	70% training, 15% validation, 15% testing
Input window	48 past observations
Forecasting horizon	24 steps ahead
Random seeds	$\{42, 123, 2024, 7, 99\}$
Primary metric	MASE
Supporting metrics	MAE, RMSE, R^2
Emulation metrics	Pearson r, KL divergence, spectral distance, autocorrelation distance, feature MSE
Lookahead prevention	Chronological split; scalers fitted on training data only
Baselines	Persistence, moving average, MLP, LSTM, Transformer, QELM teacher
CeNN variants	CeNN_NS, CeNN_Emulator

TABLE 9. Datasets used in the proof-of-concept benchmark.

Dataset	Domain	Forecasting target
Energy	Energy load forecasting	Electricity consumption/load
AAPL	Financial forecasting	Log-returns
Jena Climate	Weather forecasting	Temperature
Mackey–Glass	Chaotic dynamics	Synthetic chaotic state $x(t)$

TABLE 10. Forecasting performance of CeNN_NS using MASE.

Dataset	CeNN_NS MASE	Rank	Interpretation
Energy	0.1715	3/8	Competitive; close to MLP and LSTM
AAPL log-returns	0.0077	4/8	Difficult to interpret because naive and QELM errors are near zero
Jena Climate	0.3556	3/8	Competitive; close to MLP and LSTM
Mackey–Glass	0.0156	3/8	Strong but behind MLP and LSTM

B. FORECASTING UTILITY

Table 10 summarizes the forecasting performance of CeNN_NS using MASE as the primary metric. The results show that CeNN_NS is competitive on several datasets, but does not uniformly outperform the strongest neural baselines.

The Energy and Jena results indicate that CeNN_NS can approach the performance of strong neural baselines

FIGURE 3. MASE-based benchmark heatmap across datasets and evaluated models. Lower values indicate better forecasting performance. The figure shows that CeNN_NS is competitive, but not uniformly superior to MLP and LSTM across all datasets.

under a common chronological protocol. On Mackey–Glass, the CeNN remains highly accurate but is outperformed by MLP and LSTM. The AAPL log-return case is less informative as a forecasting benchmark because the persistence and QELM errors are nearly degenerate under the selected transformation. For this reason, AAPL is treated as a stress-test rather than as central evidence for the proposed framework.

These findings support a bounded conclusion: CeNN_NS is a competitive forecasting model under the tested conditions, but the evidence does not support a claim of universal forecasting superiority.

C. FUNCTIONAL EMULATION FIDELITY

Functional emulation was evaluated by comparing CeNN outputs with the QELM reference model. The QELM teacher provides two target types: multi-step forecasting outputs and enriched observable vectors containing $\langle Z_i \rangle$ and $\langle Z_i Z_j \rangle$. The CeNN emulation head is trained to approximate these observable-level targets.

Table 11 summarizes the main emulation-fidelity results using Pearson correlation between CeNN and QELM predictions. These results should be interpreted together with the additional emulation metrics reported in the reproducibility outputs, including KL divergence, spectral distance, autocorrelation distance, and observable-feature MSE.

The strongest functional emulation is observed on Mackey–Glass, where CeNN_Emulator reaches high alignment with the QELM reference. Energy and Jena show measurable but weaker alignment. These results indicate that QELM functional emulation is domain-dependent. They also show why the forecasting and emulation objectives must be interpreted separately: CeNN_NS is better suited to forecasting utility, while CeNN_Emulator is useful for studying QELM alignment, even when its raw forecasting accuracy is weaker.

TABLE 11. Functional emulation fidelity between CeNN and QELM reference outputs.

Dataset	CeNN_NS r	CeNN_Emulator r	Interpretation
Energy	0.3370	0.5544	Moderate alignment
AAPL log-returns	–	–	Not interpretable under near-degenerate outputs
Jena Climate	0.2213	0.3942	Weak to moderate alignment
Mackey–Glass	0.6635	0.8632	Strongest alignment

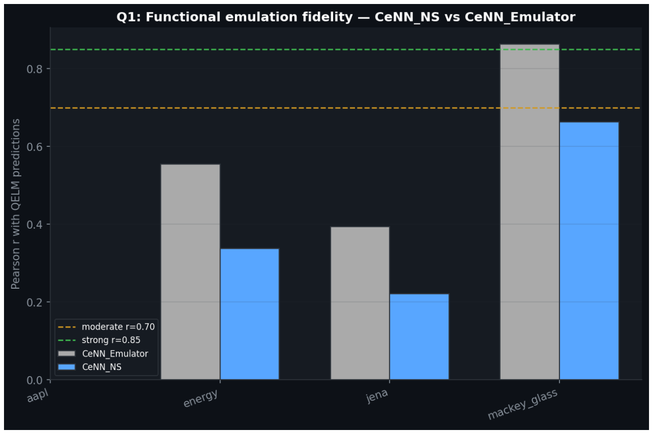

FIGURE 4. Functional emulation comparison between CeNN variants and the QELM reference model. The strongest QELM alignment is observed on the Mackey–Glass benchmark, while energy and weather series show weaker alignment.

Thus, the evidence supports functional emulation as a measurable research direction, not as a universal reproduction of QELM behavior across all domains.

D. SCALABILITY ANALYSIS

The main computational motivation for the CeNN framework is its local connectivity. Under the assumptions formalized in Section V-G, each cell interacts with a bounded number of neighbors. Therefore, the number of local interactions grows linearly with the number of cells N when the neighborhood size is fixed [16], [50].

The empirical scalability experiment varies the number of cells and measures the number of parameters, training time, and inference latency. The strongest evidence is observed for parameter growth, which follows a near-linear trend:

$$R^2_{\text{params}} = 0.9670. \tag{37}$$

This supports the architectural scaling claim of the proposed CeNN formulation.

Timing and latency measurements are less stable. They depend on the execution environment, batching, GPU scheduling, software versions, and measurement overhead. For this reason, the runtime results are re-

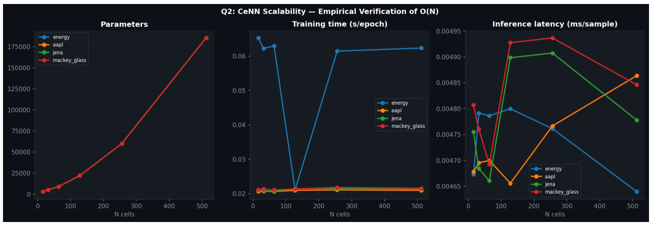

FIGURE 5. Scalability analysis of the CeNN architecture. Parameter growth supports the expected near-linear architectural trend, while timing and latency remain sensitive to implementation and hardware conditions.

TABLE 12. Ablation summary on the Energy dataset.

Variant	MASE	Interpretation
Data-only CeNN	0.1696	Best forecasting accuracy in this ablation
Balanced CeNN_NS	0.1697	Nearly equivalent forecasting accuracy with emulation terms
Dedicated CeNN_Emulator	0.7396	Stronger emulation focus but weaker forecasting accuracy

ported as implementation-level indicators rather than as hardware-independent deployment guarantees.

The computational claim must therefore be stated carefully. The CeNN does not simulate the same quantum state more efficiently than a state-vector simulator. It changes the target of approximation from quantum-state fidelity to task-relevant functional fidelity. The scalability advantage is therefore architectural and functional, not evidence of physical quantum advantage.

E. ABLATION AND COMPONENT SENSITIVITY

Ablation experiments were conducted to assess whether emulation and neuro-symbolic loss components improve forecasting accuracy or functional alignment. The results show a trade-off rather than a uniform improvement. This interpretation is consistent with the broader role of neuro-symbolic constraints as structured regularizers rather than guaranteed accuracy improvers [45], [46].

On the Energy dataset, the data-only CeNN and the balanced neuro-symbolic variant achieve very similar forecasting accuracy:

$$\mathrm{MASE}_{\text{data only}} = 0.1696, \qquad \mathrm{MASE}_{\text{balanced NS}} = 0.1697. \tag{38}$$

The dedicated emulator variant obtains weaker forecasting accuracy:

$$\mathrm{MASE}_{\text{dedicated emulator}} = 0.7396. \tag{39}$$

These results show that the emulation-oriented components should not be interpreted as universally improving raw forecasting accuracy. Their main role is to expose and quantify the trade-off between direct predictive performance and QELM functional alignment.

This ablation narrows the claim of the framework. The neuro-symbolic and emulation terms are not presented as guaranteed accuracy boosters. They are used to formalize, measure, and study functional alignment with a quantum-inspired reference model.

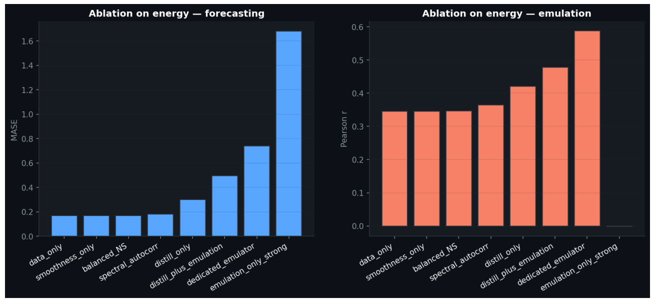

FIGURE 6. Ablation overview illustrating the trade-off between forecasting accuracy and emulation-oriented objectives. The results should be interpreted as component sensitivity rather than proof that all neuro-symbolic terms improve forecasting accuracy.

F. STATISTICAL COMPARISON

Statistical comparisons were performed across the five random seeds for trainable models. The purpose was to assess whether the observed differences between CeNN_NS and neural baselines are stable across runs. The comparison used paired tests and effect-size indicators when enough paired values were available, including Wilcoxon signed-rank testing and effect-size interpretation [51], [52].

The results confirm that CeNN_NS is competitive but not uniformly superior. On Energy and Jena, its mean MASE is close to MLP and LSTM. On Mackey–Glass, MLP and LSTM remain stronger. On AAPL log-returns, the near-zero errors of simple or QELM-based baselines make the comparison less informative.

Because only five seeds are used, statistical tests are interpreted cautiously. The tests support the descriptive conclusion that CeNN is a competitive model under the selected protocol, but they do not justify a general claim of dominance over neural baselines.

G. DISCUSSION OF FINDINGS

The proof-of-concept benchmark supports four findings.

First, CeNN_NS achieves competitive forecasting performance on multiple datasets under a common chronological evaluation protocol. This supports the feasibility of local CeNN dynamics as a forecasting substrate.

Second, functional emulation is measurable but domain-dependent. The strongest QELM alignment is observed on Mackey–Glass, while Energy and Jena show weaker alignment. This suggests that CeNN-based functional emulation is not a universal property; it depends on the dataset, reference teacher, and target dynamics.

Third, the scalability analysis supports near-linear architectural growth in the number of CeNN parameters. However, runtime and latency measurements remain hardware-sensitive and should not be used alone to claim deployment readiness.

Fourth, the ablation results show a trade-off between forecasting accuracy and emulation alignment. The full neuro-symbolic formulation is useful for studying this trade-off, but its components should not be assumed to improve raw forecasting accuracy in every domain.

Overall, the benchmark strengthens the empirical basis of the paper by providing a reproducible, multi-domain, baseline-aware evaluation. The results support CeNN-based functional emulation as a tractable research direction for scalable quantum-inspired TSF, while also defining its current limits.

VII. CRITICAL APPRAISAL AND OPEN CHALLENGES

The synthesis of the reviewed QML-TSF literature reveals a persistent gap between the representational promise of quantum and quantum-inspired models and the methodological requirements of reliable forecasting. While quantum feature maps, variational circuits, kernels, and reservoir-like dynamics can provide expressive modeling mechanisms [5]–[8], [18], [19], the reviewed corpus also shows recurring limitations in scalability, benchmarking fairness, reproducibility, hardware realism, and claim calibration [12], [14], [15].

This section critically discusses these limitations and identifies the conditions under which QML-TSF research can become more reliable, comparable, and reproducible. The objective is not to restate the taxonomy or the proof-of-concept results, but to extract the main methodological lessons that motivate careful reporting and bounded interpretation.

A. SYSTEMIC LIMITATIONS OF CURRENT QML-TSF RESEARCH

Three limitations recur across the reviewed literature: incomplete reproducibility, heterogeneous benchmarking, and a disconnect between simulation-based results and hardware-aware validation.

1) Reproducibility and Parameter Opacity

As shown in the risk-of-bias assessment in Table 3, reproducibility is one of the weakest dimensions of the reviewed corpus. This observation is consistent with broader concerns in systematic-review methodology and risk-of-bias assessment, where incomplete reporting limits the interpretability and replicability of empirical claims [17], [53]. Only 42% of the included studies were rated as low risk in terms of data, code, and experimental reproducibility. This limitation appears through missing source code, incomplete preprocessing descriptions, unavailable train-validation-test splits, unreported random seeds, and insufficient configuration details.

This gap is particularly important in QML-TSF because performance may depend strongly on encoding strategy, circuit depth, ansatz structure, optimizer choice, initialization, shot count, simulator backend, or noise model. In variational settings, these factors can also affect trainability through phenomena such as barren plateaus [14]. Without transparent reporting, it becomes difficult to determine whether reported gains are due to the quantum or quantum-inspired component, the preprocessing pipeline, baseline under-optimization, or favorable dataset selection.

2) Benchmarking Heterogeneity

A second limitation is the absence of standardized benchmarking protocols. The reviewed studies vary substantially in datasets, forecasting horizons, temporal splits, evaluation metrics, baseline selection, and tuning procedures. As a result, performance claims are often difficult to compare across studies.

Two issues are central. First, several studies compare QML models against limited or insufficiently tuned baselines. Fair TSF evaluation requires not only simple statistical baselines, but also competitive machine learning and deep learning baselines such as LSTM, Transformer-type models, N-BEATS, or Autoformer-like architectures where appropriate [3], [4], [12], [54], [55]. Second, studies often report different metrics, such as RMSE, MAE, MAPE, sMAPE, or R^2, without confidence intervals, statistical tests, or effect sizes. This limits the possibility of strong cross-study conclusions and supports a cautious interpretation of reported advantages. Metrics such as MASE are particularly useful when comparison against naive forecasting references is required [49].

3) Simulation–Hardware Disconnect

A third limitation concerns the gap between idealized simulation and physical quantum execution. Many QML-TSF experiments rely on small simulated circuits, often with limited qubit counts and simplified or absent noise models. Such experiments are useful for early-stage exploration, but they do not fully represent the constraints of current NISQ devices.

Noise-free state-vector simulations may overestimate performance relative to physical execution, where decoherence, gate errors, finite-shot effects, measurement noise, and restricted connectivity affect results [14], [15]. At the same time, state-vector simulation itself scales exponentially with the number of qubits, limiting its use for larger or higher-dimensional forecasting problems. Therefore, claims of practical QML advantage should be made only under hardware-aware, reproducible, and baseline-fair evaluation protocols.

B. GOVERNANCE, INTERPRETABILITY, AND ROBUSTNESS

Forecasting models used in domains such as energy, finance, transportation, and healthcare may influence operational, economic, or safety-related decisions. For this reason, model evaluation should not be limited to error metrics. Interpretability, auditability, robustness, and uncertainty assessment are also important.

Many QML and hybrid models operate through high-dimensional quantum states, kernels, or latent representations that are difficult to inspect directly. This creates an explainability challenge in regulated or safety-critical settings. Neuro-symbolic approaches may partially address this issue by introducing explicit constraints, structured penalties, or domain rules into the learning process [45], [46], [56]. However, such mechanisms do not make a model fully explainable by themselves. They should be combined with uncertainty quantification, robustness testing, feature attribution, residual analysis, and failure-mode assessment.

Distribution shift is another critical challenge. Energy demand, financial markets, mobility patterns, and physiological signals can change over time. Future QML-TSF studies should therefore evaluate models not only under standard chronological splits, but also under regime shifts, missing data, noise perturbations, and out-of-distribution conditions. This is especially important when high-dimensional quantum or quantum-inspired transformations may obscure how input biases are transformed internally. Robustness under distribution shift is a central concern in modern machine learning and forecasting systems [1], [57].

C. ACCESSIBILITY AND RESOURCE REQUIREMENTS

Access to quantum hardware, specialized simulators, and high-performance computing resources remains uneven across institutions. This can limit independent verification and reduce participation in QML-TSF research. Quantum-inspired functional emulation on classical hardware can partially reduce this barrier by enabling controlled experiments without immediate access to quantum processing units.

This accessibility benefit should not be confused with physical quantum execution. Functional emulation is a complementary pathway for studying selected task-relevant behaviors associated with quantum-inspired models. It improves experimental accessibility and reproducibility, but it does not replace hardware validation when claims concern physical quantum computation. Open-source quantum and scientific-computing ecosystems can support this accessibility objective, but only if studies release sufficient code, configuration, and output metadata for independent inspection [58]–[60].

D. RECOMMENDED REPORTING PROTOCOL FOR QML-TSF STUDIES

To improve reliability and comparability, future QML-TSF studies should report a minimum set of methodological details. The following checklist is proposed as a practical reporting protocol rather than as a universal standard:

1) **Model architecture:** report the complete architecture, including circuit topology when applicable, encoding strategy, qubit count, circuit depth, ansatz type, number of trainable parameters, reservoir configuration, or emulator design.
2) **Execution modality:** clearly state whether the model is evaluated through state-vector simulation, density-matrix simulation, quantum-inspired functional emulation, or physical quantum hardware execution.
3) **Baseline hierarchy:** include appropriate statistical, machine learning, and deep learning baselines. Where relevant, this should include ARIMA or seasonal naive models, tree-based methods, LSTM-type models, Transformer-type models, and other competitive TSF architectures [1], [3], [4], [54], [55].
4) **Evaluation metrics and uncertainty:** report multiple forecasting metrics together with standard deviations, confidence intervals, statistical tests, or effect sizes when possible [49], [51], [52].
5) **Computational profiling:** report training time, inference latency, memory footprint, number of trainable parameters, shot count where applicable, simulator or hardware backend, and software versions.
6) **Noise and robustness testing:** for simulated or hardware quantum experiments, report noise assumptions and, where possible, evaluate robustness under depolarizing noise, phase damping, finite-shot effects, or backend-specific noise models [14], [15].
7) **Reproducibility package:** provide datasets or access instructions, preprocessing scripts, chronological splits, random seeds, configuration files, source code, and output tables whenever licensing and ethical constraints permit [12], [17].

E. IMPLICATIONS FOR FUNCTIONAL EMULATION

The gaps identified above motivate the exploration of quantum-inspired functional emulation. The motivation is not that emulation replaces quantum hardware or reproduces quantum mechanics. Rather, functional emulation provides a controlled classical setting in which selected forecasting-relevant behaviors associated with quantum-inspired reference models can be approximated, measured, and compared under reproducible conditions.

The CeNN framework introduced in Section V is one bounded response to these gaps. Its local dynamical structure addresses architectural scalability under explicit bounded-neighborhood assumptions [16], [50], while its emulation and regularization objectives provide a way to study the trade-off between forecasting accuracy and functional alignment. However, the framework should not be interpreted as a definitive solution to QML-TSF scalability, interpretability, or deployment.

Overall, progress in QML-TSF will depend less on broad claims of quantum advantage and more on disciplined methodology: transparent reporting, fair baselines, hardware-aware evaluation, reproducible code, and a clear distinction between physical quantum computation, classical quantum simulation, and quantum-inspired functional emulation.

VIII. LIMITATIONS AND PATH TO HARDWARE-AWARE VALIDATION

The proposed CeNN framework is a classical quantum-inspired functional emulator. It is designed to approximate selected task-relevant behaviors associated with a QELM-style reference model, but it is not a substitute for physical quantum computation. This distinction is central to the scope of the paper. The framework does not reproduce full quantum-state evolution, genuine entanglement, phase-coherent interference, or physical quantum computational advantage [6], [7], [14].

This section summarizes the main limitations of the proposed approach and outlines a cautious path toward hardware-aware validation. The objective is not to claim that CeNN can replace quantum processors, but to clarify how classical functional emulation can support reproducible benchmarking, controlled experimentation, and model development while current quantum hardware remains constrained by NISQ-era limitations [14], [15].

A. FUNCTIONAL FIDELITY VERSUS PHYSICAL FIDELITY

A central limitation of the proposed framework is that it targets functional fidelity, not physical fidelity. Functional fidelity refers to the approximation of selected output distributions, forecasting responses, spectral structures, temporal-correlation profiles, and observable-level behaviors. Physical fidelity, by contrast, requires reproducing the underlying quantum-state evolution or measurement statistics of a quantum system.

The CeNN framework belongs to the first category. It can be used to study whether selected quantum-inspired forecasting behaviors can be approximated by a scalable classical dynamical model, but it cannot support claims about physical quantum dynamics. In particular, the proposed CeNN:

- does not represent an n-qubit state vector or density matrix;
- does not reproduce genuine quantum entanglement;
- does not implement unitary quantum evolution;
- does not demonstrate quantum computational advantage;
- does not validate deployment on a physical quantum processing unit.

The results should therefore be interpreted as evidence for a classical functional-emulation pathway. They do not establish that the proposed model physically realizes quantum computation or reproduces quantum-state evolution.

B. SCOPE OF THE EMPIRICAL EVIDENCE

The proof-of-concept benchmark was designed to be transparent and reproducible, but it remains bounded. The experimental protocol covers four datasets, a fixed chronological splitting strategy, a 48-step input window, a 24-step forecasting horizon, and five random seeds. This design enables controlled comparison across baselines, but it does not establish generality across all forecasting domains.

The results show domain-dependent behavior. CeNN_NS achieves competitive forecasting performance under the tested protocol, but it does not uniformly outperform MLP or LSTM. Functional emulation is strongest on the Mackey–Glass chaotic benchmark and weaker on energy and weather series. The AAPL log-return case is less informative as a central forecasting benchmark because some reference and naive outputs become nearly degenerate under the selected transformation.

These findings support the feasibility of CeNN-based functional emulation, but they do not justify broad claims of universal superiority, operational deployment, or general replacement of classical or quantum-inspired forecasting models. Broader validation is required on additional datasets, multivariate time series, longer horizons, noisy settings, and regime-shift scenarios. Such extensions are especially important because forecasting models can be sensitive to non-stationarity, distribution shift, and evaluation protocol design [1], [48], [57].

C. TEACHER DEPENDENCE AND EMULATION CALIBRATION

The functional-emulation component depends on the quality of the QELM reference model. QELM-style models are attractive because they combine fixed or quantum-inspired feature mappings with efficient classical readouts, but their performance can depend strongly on the feature map, random projection mechanism, observable construction, and regularization strategy [23], [24].

If the teacher is weak, unstable, or poorly calibrated, imitating it may not improve forecasting accuracy and may introduce an unhelpful training signal. For this reason, the framework includes a teacher-quality control mechanism that reduces the influence of the QELM teacher when it performs poorly relative to a naive validation baseline.

This mechanism improves methodological caution, but it also reveals a limitation: QELM alignment is not automatically beneficial. The ablation results show that emulation-oriented terms and neuro-symbolic regularization should not be interpreted as universal accuracy boosters. Their role is to formalize and measure the trade-off between forecasting performance and functional alignment with the reference model.

Future work should investigate stronger QELM teachers, ensembles of quantum-inspired reference models, adaptive loss weighting, and dataset-specific calibration of emulation, spectral, autocorrelation, and regularization terms.

D. SCALABILITY AND RUNTIME LIMITATIONS

The main scalability argument of the CeNN framework is architectural. Under bounded-neighborhood assumptions, each cell interacts with a fixed number of neighbors, so the number of local interactions grows linearly with the number of cells. This follows the classical CeNN principle of locally coupled cellular dynamics [16], [44], [50].

However, architectural scaling should not be confused with complete runtime scaling. Training time and inference latency depend on implementation details, batching, GPU scheduling, software versions, memory layout, and hardware conditions. In the current experiments, timing and latency measurements are therefore treated as implementation-level indicators rather than hardware-independent deployment guarantees.

The CeNN also does not simulate the same quantum state more efficiently than a state-vector simulator. Its computational advantage arises from changing the approximation target: it replaces state-vector fidelity with selected task-level functional fidelity. This is a modeling and evaluation choice, not a claim of physical quantum acceleration.

E. PATH TO HARDWARE-AWARE VALIDATION

The path from quantum-inspired functional emulation to hardware-aware QML validation should proceed in stages.

First, future experiments should compare CeNN-based emulation with noise-aware simulations of QML reference models. These simulations should include perturbations such as finite-shot sampling, depolarizing noise, phase damping, amplitude damping, or backend-specific noise models where available. Noise-aware evaluation is important because idealized simulations can overestimate performance relative to NISQ execution [14], [15], [61].

A generic robust-training objective can be written as

$$\mathcal{L}_{\text{robust}}(\theta) = \mathbb{E}_{\eta \sim \mathcal{N}} \left[\mathcal{L}_{\text{total}}(\theta; \eta) \right], \tag{40}$$

where η denotes noise parameters sampled from a distribution $\mathcal{N}$, and $\mathcal{L}_{\text{total}}$ is the training objective defined in Section V. This formulation is a future robustness extension, not a claim that the current CeNN implementation is already transferable to quantum hardware.

Second, hardware-aware validation should compare three execution modalities: state-vector or density-matrix simulation, quantum-inspired functional emulation, and physical quantum execution. The comparison must use matched data splits, preprocessing, metrics, and baselines. Without this alignment, differences between models may reflect experimental design rather than genuine modeling behavior.

Third, once suitable QPU access is available, QELM or related QML reference models should be executed on physical hardware and compared against the CeNN emulator using the same functional metrics used in this paper: forecasting error, output-distribution alignment, spectral distance, autocorrelation distance, and observable-level agreement.

F. INTERPRETABILITY AND GOVERNANCE LIMITS

The neuro-symbolic component of the framework supports regularization and partial auditability, but it does not provide full explainability. Symbolic or differentiable constraints can guide the model toward smoother or more domain-consistent predictions, but they do not establish causal interpretability or complete transparency [45], [46], [56].

In high-stakes domains such as energy, finance, healthcare, and transportation, additional governance mechanisms would be required before deployment. These include uncertainty quantification, residual analysis, robustness testing, feature attribution, failure-mode analysis, monitoring under distribution shift, and domain-specific validation.

Thus, the neuro-symbolic layer should be interpreted as one component of a broader auditability strategy, not as a complete solution to explainability, robustness, or governance.

G. SUMMARY

The proposed CeNN framework provides a computationally tractable and reproducible environment for studying quantum-inspired functional emulation in TSF. Its value lies in testing whether selected forecasting-relevant behaviors of a QELM-style reference model can be approximated by a locally connected classical dynamical architecture.

The limitations are equally clear. The framework does not establish physical quantum equivalence, hardware readiness, or quantum advantage. Its empirical evidence remains domain-dependent, its emulation quality depends on the reference teacher, and its timing behavior is implementation- and hardware-sensitive. Future progress requires broader datasets, hardware-aware simulations, stronger QELM references, direct comparison with physical QML implementations when feasible, and more systematic robustness analysis.

Until such evidence is obtained, CeNN should be interpreted as a bounded methodological bridge for scalable quantum-inspired forecasting research, not as

TABLE 13. Computational environment and reproducibility metadata.

Component	Configuration
Execution environment	Cloud notebook environment with GPU support
Programming language	Python scientific-computing stack
Main ML framework	PyTorch [60]
Numerical libraries	NumPy, Pandas, SciPy, scikit-learn [62]–[65]
Visualization	Matplotlib [66]
Execution backend	Standard tensor operations; no custom CUDA kernel
Metadata export	Software versions, device information, configuration, and output paths stored in `metadata_full.json`

a replacement for quantum hardware or a deployment-ready forecasting system.

IX. REPRODUCIBILITY AND IMPLEMENTATION DETAILS

To address the reproducibility concerns identified in the systematic review, this section reports the implementation details of the proof-of-concept benchmark. The objective is not to claim bitwise-identical reproducibility across all computational environments, but to provide sufficient information for controlled replication within reasonable numerical tolerance.

The implementation was developed as a reproducibility-oriented research prototype using standard open-source scientific-computing tools. The main software stack includes Python, PyTorch, NumPy, Pandas, SciPy, scikit-learn, and Matplotlib [60], [62]–[66]. This choice was intentional: the benchmark is designed to be inspectable and reusable without custom low-level kernels or proprietary execution environments.

A. COMPUTATIONAL ENVIRONMENT

The experiments were conducted in a cloud notebook environment with GPU acceleration enabled. Tensor operations for the trainable neural baselines, the QELM teacher, and the CeNN variants were executed through the standard PyTorch backend when GPU acceleration was available [60]. No custom CUDA kernel was required.

Because cloud notebook environments may vary across sessions, runtime and latency measurements are interpreted as environment-specific indicators rather than hardware-independent deployment guarantees. The reported timing values support comparative profiling under the same execution environment, not production-level performance claims.

B. SOFTWARE STACK AND REPOSITORY ORGANIZATION

The codebase separates dataset loading, preprocessing, model definition, training, evaluation, statistical analysis, visualization, and export of results. This organization directly addresses the reproducibility gap identified in the reviewed QML-TSF literature.

The main components are:

- **Dataset utilities:** loading or generation of the four benchmark datasets, including deterministic generation for Mackey–Glass dynamics [47].
- **Preprocessing utilities:** chronological filtering, 70/15/15 chronological splitting, train-only normalization, sliding-window construction, and leakage prevention.
- **Baseline implementations:** persistence, moving average, MLP, LSTM, Transformer, and QELM teacher models.
- **CeNN implementation:** locally connected CeNN core, forecasting head, emulation head, and loss components for forecasting, distillation, observable emulation, spectral alignment, autocorrelation alignment, and smoothness regularization.
- **Training and evaluation routines:** early stopping, multi-seed execution, test-set prediction, metric computation, functional-emulation scoring, latency measurement, and result logging.
- **Statistical analysis utilities:** paired run-level comparisons, confidence intervals, paired t-tests, Wilcoxon signed-rank tests, and effect-size reporting [51], [52].
- **Export utilities:** automatic generation of CSV files, metadata files, figures, tables, and reproducibility outputs.

The public repository contains the main notebook, dependency files, generated outputs, figures, metadata, and documentation. The repository is available through the reproducibility package link reported in the Data and Code Availability section.

C. DATASETS AND CHRONOLOGICAL SPLITTING

The proof-of-concept benchmark uses four datasets covering energy load forecasting, financial log-return forecasting, weather temperature forecasting, and deterministic chaotic dynamics. Three datasets correspond to real-world time series, while Mackey–Glass is used as a controlled chaotic benchmark [47].

For each dataset, up to 20 000 chronological observations were used after cleaning and transformation. No random shuffling was applied. Each series was split chronologically: the first 70% of observations were used for training, the next 15% for validation, and the final 15% for testing. Normalization parameters were fitted only on the training split and then applied to the validation and test splits. This prevents lookahead bias, which is a central concern in time-series forecasting [1], [48].

TABLE 14. Data splitting and preprocessing protocol.

Item	Protocol
Observations	Up to 20 000 chronological observations per dataset
Split type	Strict chronological split
Training set	First 70% of observations
Validation set	Next 15% of observations
Test set	Final 15% of observations
Input window	48 past observations
Forecasting horizon	24-step forecasting horizon
Normalization	Scaler fitted on training data only
Windowing	Sliding-window construction using past observations only
Leakage prevention	No future information used in preprocessing, scaling, or training

D. RANDOM SEEDS AND EXPERIMENTAL REPETITION

Trainable models were evaluated over five fixed random seeds:

$$\mathcal{S} = \{42, 123, 2024, 7, 99\}. \quad (41)$$

The seeds were used to initialize model parameters, data-loader behavior where applicable, and numerical libraries where supported. Results are reported as mean and standard deviation across runs when repeated-run outputs are available.

Because GPU operations and cloud notebook environments may introduce residual nondeterminism, the goal is controlled reproducibility rather than bitwise-identical reproduction. This distinction is important for neural and hybrid forecasting experiments, where backend scheduling and floating-point operations can produce small numerical differences even under fixed seeds.

E. MODEL FAMILIES AND BASELINES

The benchmark includes naive, neural, quantum-inspired, and CeNN-based models:

- **Persistence:** repeats the last observed value across the forecast horizon.
- **Moving average:** repeats the mean of the input window.
- **MLP:** feed-forward neural baseline trained on sliding-window inputs.
- **LSTM:** recurrent neural baseline for temporal dependency modeling [3].
- **Transformer:** attention-based neural baseline inspired by sequence modeling architectures [4].
- **QELM teacher:** quantum-inspired reference model used to produce prediction targets and enriched observable features [23], [24].
- **CeNN_NS:** balanced CeNN model optimized for forecasting utility and partial functional alignment.
- **CeNN_Emulator:** emulation-oriented CeNN variant used to study alignment with QELM predictions and observables.

TABLE 15. Main hyperparameters used in the proof-of-concept benchmark.

Hyperparameter	Value
Optimizer	Adam [67]
Input window length	48
Forecasting horizon	24
Batch size	256
Maximum epochs	80
Early stopping patience	12 epochs
QELM qubits	4
QELM layer search	1–4 layers
QELM ridge regularization	Selected on validation grid
CeNN cells	128
CeNN layers	4
CeNN kernel size	5
CeNN topology	One-dimensional circular local connectivity
CeNN variants	CeNN_NS and CeNN_Emulator
Scalability cells	16, 32, 64, 128, 256, 512

All directly compared models use the same chronological splits, preprocessing logic, input window, forecasting horizon, and test sets. This reduces the risk that performance differences arise from inconsistent data handling rather than model behavior.

F. MODEL CONFIGURATION AND HYPERPARAMETERS

The main hyperparameters used in the benchmark are reported in Table 15. Values were fixed across datasets unless otherwise stated. More detailed configuration information is provided in the exported metadata file and repository configuration files.

The CeNN configuration uses local circular interactions rather than a dense recurrent matrix. With a fixed kernel size, the number of local interactions grows linearly with the number of cells. This implementation matches the local-connectivity assumptions used in the formal complexity analysis.

The QELM teacher is selected using the validation split. If its validation performance is weaker than a naive reference, its influence is reduced in the CeNN training objective. This prevents the CeNN from being forced to imitate an uninformative teacher.

G. EVALUATION METRICS AND STATISTICAL ANALYSIS

Forecasting performance was evaluated primarily using Mean Absolute Scaled Error (MASE), which compares model error against a naive forecast and is more stable than percentage-based metrics when target values can approach zero [49]. Supporting metrics include MAE, RMSE, and R^2.

Functional emulation was evaluated using Pearson correlation between CeNN and QELM predictions, KL divergence between output distributions, spectral distance, autocorrelation-profile distance, mean-squared error between CeNN emulation outputs and QELM observable features, and mean observable correlation. These metrics assess functional alignment, not physical quantum-state fidelity.

Computational behavior was assessed using parameter count, training time, and inference latency. Timing values are reported as implementation-level indicators and should not be interpreted as hardware-independent deployment guarantees.

For statistical reliability, run-level paired comparisons were computed across the five seeds for trainable models. The analysis reports paired t-test values, Wilcoxon signed-rank results, confidence intervals, and effect-size indicators where applicable [51], [52]. Because the number of seeds is limited, statistical tests are interpreted cautiously and jointly with effect magnitude and dataset-level consistency.

H. SCALABILITY AND MEMORY PROFILING

The scalability analysis focuses on the local-connectivity structure of the CeNN. Under a fixed neighborhood size, each cell interacts with a bounded number of neighbors. Therefore, the number of local interactions grows linearly with the number of CeNN cells, as formalized in Section V-G.

The empirical scalability experiment varied the number of CeNN cells across six values:

$$N \in \{16, 32, 64, 128, 256, 512\}.$$

The analysis measured parameter count, training time, and inference latency. The strongest empirical support was obtained for near-linear parameter growth. Training time and latency are reported, but interpreted cautiously because they depend on batching, GPU scheduling, software versions, memory layout, and timing procedure.

Larger CeNN configurations correspond to more classical CeNN cells, not to more qubits. A CeNN cell is a classical dynamical unit and should not be interpreted as a quantum bit.

I. REPRODUCIBILITY PACKAGE

The reproducibility package accompanying this manuscript is intended to support independent inspection and controlled replication of the proof-of-concept experiments. The repository contains the main notebook, requirements file, generated outputs, figures, metadata, and documentation.

The main reproducibility files include:

- `summary_full.csv`;
- `benchmark_full.csv`;
- `emulation_summary_full.csv`;
- `emulation_fidelity_full.csv`;
- `ablation_full.csv`;
- `scalability_full.csv`;
- `linear_fit_full.csv`;
- `statistical_comparison_full.csv`;
- `metadata_full.json`;
- exported figures for forecasting, emulation, ablation, teacher selection, scalability, and architecture visualization.

The repository is designed to make the experiments auditable without requiring the reader to inspect an opaque compressed file. The exported metadata and CSV files allow independent verification of the reported tables, figures, and conclusions.

J. REPRODUCIBILITY BOUNDARIES

Several boundaries must be acknowledged. First, cloud notebook environments may allocate different hardware or runtime conditions across sessions, which can affect timing measurements. Second, GPU operations may introduce small nondeterministic numerical differences even when random seeds are fixed. Third, exact reproduction depends on dataset versions, package versions, backend behavior, and hardware allocation.

For these reasons, the goal is not bitwise-identical reproduction, but transparent and controlled replication. The reporting of dataset sources, chronological splits, preprocessing rules, seeds, software dependencies, configuration files, result tables, figures, and metadata is intended to make the experiments auditable and repeatable within reasonable numerical tolerance.

K. SUMMARY

The implementation was designed as a reproducibility-oriented proof of concept. It reports the computational environment, software stack, dataset handling, chronological splits, random seeds, model families, hyperparameters, evaluation metrics, statistical tests, scalability assumptions, exported outputs, and reproducibility boundaries.

This section directly addresses one of the main weaknesses identified in the reviewed QML-TSF literature: incomplete reporting of code, preprocessing, seeds, configuration files, and execution environment. The resulting reproducibility package provides a transparent basis for further validation, extension to additional datasets, and comparison against stronger classical, quantum-inspired, and hardware-executed QML baselines.

X. FUTURE DIRECTIONS AND RESEARCH AGENDA

The results and synthesis presented in this paper suggest that QML and quantum-inspired methods remain promising but methodologically immature for time series forecasting (TSF). Current evidence supports further investigation rather than immediate claims of general superiority, physical quantum advantage, or industrial readiness. Future progress will require stronger theoretical foundations, standardized benchmarks, reproducible implementations, hardware-aware evaluation, and careful integration with domain constraints [5]–[7], [12].

This section outlines a research agenda for advancing QML-TSF from exploratory proof-of-concept studies toward more reliable, comparable, and deployment-oriented forecasting research.

A. THEORETICAL FOUNDATIONS AND FORMAL GUARANTEES

A central open challenge is the gap between empirical performance and theoretical understanding. Many QML-TSF studies report encouraging results, but the conditions under which quantum, hybrid, or quantum-inspired models outperform strong classical baselines remain insufficiently characterized [12], [13], [24].

Future work should distinguish more carefully between three levels of approximation:

- **State-level fidelity:** approximation of the full quantum state or density matrix.
- **Measurement-level fidelity:** approximation of selected measurement distributions or observable statistics.
- **Task-level functional fidelity:** approximation of forecasting-relevant behaviors such as output distributions, spectral structure, temporal correlations, and predictive accuracy.

The CeNN framework belongs to the third category. Therefore, future theoretical work should derive tighter links between distributional alignment, spectral similarity, temporal-correlation preservation, and forecasting error. Such results would make the notion of functional emulation more precise and would help determine when a classical emulator can serve as a useful approximation to a quantum-inspired forecasting module.

B. OPTIMIZATION STABILITY AND TRAINABILITY

Hybrid QML models and neuro-symbolic emulators often involve non-convex optimization landscapes. Variational quantum circuits may suffer from barren plateaus, sensitivity to initialization, and finite-shot noise, while CeNN-based models may be sensitive to grid size, connectivity radius, integration depth, and auxiliary loss weights [14], [15], [68]. Future research should therefore examine:

- convergence behavior under different initialization schemes;
- sensitivity to emulation, spectral, autocorrelation, and neuro-symbolic loss weights;
- stability under noisy gradients, finite-shot sampling, and backend-dependent perturbations;
- robustness under non-stationarity, missing data, and regime shifts in temporal data.

Such analysis is necessary before QML-TSF models or functional emulators can be considered reliable beyond controlled benchmark settings.

C. EMULABILITY OF QUANTUM-INSPIRED FORECASTING MODELS

A more systematic theory of emulability is needed. Not all quantum or quantum-inspired models are equally suitable for classical functional emulation. Future work should characterize which classes of quantum circuits, reservoirs, kernels, or fixed quantum-inspired transformations can be approximated efficiently by classical dynamical systems without attempting full state-vector simulation.

This research direction should avoid broad claims that all useful quantum behaviors can be classically emulated. Instead, it should identify specific forecasting-relevant behaviors that can be approximated under explicit assumptions. For example, QELM and QRC models may be more naturally suited to functional emulation because they already rely on fixed or partially fixed feature-generating dynamics with trainable classical readouts [18], [19], [23]–[25].

D. COMPUTATIONAL INNOVATIONS AND SCALABILITY

Scalability remains a major bottleneck for QML-TSF research. State-vector simulation scales exponentially with the number of qubits, while physical quantum hardware remains constrained by NISQ noise, limited connectivity, and finite-shot uncertainty [14], [15]. Future computational work should improve scalability without obscuring the distinction between physical quantum execution, classical quantum simulation, and quantum-inspired functional emulation.

Several directions are especially relevant:

- **Tensor-network representations:** Matrix Product States, Tree Tensor Networks, and related structures may help represent selected quantum-inspired correlations with reduced memory cost when the entanglement structure is limited [27].
- **Sparse and structured operators:** Temporal, spectral, or graph-domain sparsity may reduce computational cost in long-sequence or multivariate forecasting tasks.
- **Reservoir-style architectures:** QRC, QELM, EQELM, and CeNN-like models can reduce training complexity by using fixed or locally structured dynamics with trainable readouts [16], [18], [19], [23], [25], [50].
- **Noise-aware training:** Models intended for hardware-aware claims should be evaluated under finite-shot effects, depolarizing noise, phase damping, and backend-specific noise profiles [14], [15], [61].
- **Strong classical baselines:** Any proposed quantum or quantum-inspired method should be compared against competitive classical models, including LSTM, Transformer-type architectures, N-BEATS, and Autoformer-like models where appropriate [3], [4], [54], [55].

E. TOWARD A FULL-STACK EVALUATION FRAMEWORK

Future QML-TSF research would benefit from a full-stack evaluation framework connecting hardware assumptions, algorithmic design, software implementation, and domain-specific forecasting requirements. Such a framework should include four levels:

- **Hardware and backend level:** specify whether the experiment uses state-vector simulation, density-matrix simulation, quantum-inspired functional emulation, or physical QPU execution.
- **Algorithmic level:** report the encoding strategy, circuit depth, ansatz design, reservoir configuration, emulator structure, number of parameters, and training protocol.
- **Software and reproducibility level:** release code, configurations, seeds, preprocessing scripts, environment files, backend versions, and generated outputs [17], [60], [62]–[66].
- **Application level:** use domain-appropriate datasets, forecasting horizons, chronological train-validation-test splits, metrics, uncertainty estimates, and operational constraints [1], [48], [49].

This full-stack view is important because many reported QML-TSF gains may depend as much on preprocessing, baseline selection, tuning, and evaluation protocol as on the quantum or quantum-inspired component itself.

F. PROPOSED RESEARCH AGENDA

Table 16 summarizes a cautious research agenda for QML-TSF. The timeline is indicative rather than predictive. Its purpose is to organize priorities, not to assert that specific deployment milestones will necessarily be achieved by a given date.

This agenda avoids treating quantum advantage or industrial deployment as immediate outcomes. Instead, it emphasizes the intermediate methodological steps required before such claims can be responsibly made.

G. RECOMMENDATIONS FOR IMMEDIATE ACTION

Based on the review findings and the proof-of-concept benchmark, the following actions would improve the reliability and comparability of future QML-TSF research:

1) **Develop standardized QML-TSF benchmarks:** define shared datasets, temporal splits, forecasting horizons, preprocessing rules, and evaluation metrics for domains such as energy, finance, weather, traffic, and controlled chaotic systems.
2) **Report stronger baseline comparisons:** compare quantum and quantum-inspired models against statistical baselines, machine learning models, and modern deep learning baselines such as LSTM, Transformer-type models, N-BEATS, and Autoformer-like architectures [3], [4], [54], [55].

TABLE 16. Indicative research agenda for QML-TSF.

Pillar	Horizon	Objective and evaluation focus
Standardized benchmarking	Short	Shared datasets, chronological splits, common baselines, and reproducible leaderboards.
Reproducibility protocols	Short	Transparent reporting of code, seeds, preprocessing, circuit parameters, backends, and generated outputs.
Functional emulation validation	Short–medium	Assess whether emulators such as CeNN preserve output distributions, observable statistics, spectra, temporal correlations, and forecasting accuracy.
Hardware-aware evaluation	Medium	Compare ideal simulation, noisy simulation, functional emulation, and physical QPU execution under matched datasets and metrics.
Robustness and uncertainty	Medium	Test models under noise, missing data, distribution shift, and regime changes using robustness and calibration metrics.
Deployment-oriented studies	Long	Evaluate latency, reliability, interpretability, governance, monitoring, and domain-specific validation in realistic workflows.

Note: This agenda is indicative and summarizes key priorities for improving reproducibility, scalability, hardware awareness, and deployment readiness in QML-TSF research.

3) **Improve reproducibility packages:** provide source code, random seeds, configuration files, dataset versions, preprocessing scripts, environment specifications, hardware or simulator metadata, and generated output tables [12], [17].
4) **Separate execution modalities:** clearly distinguish between state-vector simulation, noisy simulation, quantum-inspired functional emulation, and physical quantum hardware execution.
5) **Include hardware-aware noise tests:** when hardware transfer is part of the claim, evaluate models under realistic noise assumptions, finite-shot sampling, and backend-specific constraints [14], [15].
6) **Assess robustness beyond average accuracy:** report performance under regime shifts, missing data, distribution shift, adversarial perturbations, and extreme events [1], [57].
7) **Clarify the role of neuro-symbolic constraints:** when symbolic or differentiable rules are used, report whether they improve accuracy, robustness, interpretability, or domain consistency, rather than assuming that they improve all dimensions simultaneously [45], [46].

H. SYNTHESIS

The future of QML-TSF depends on methodological discipline. The field will progress less through broad claims of quantum advantage and more through reproducible experiments, strong baselines, hardware-aware evaluation, and clear definitions of what is being simulated, emulated, or physically executed.

For the proposed CeNN framework, the most important next steps are broader dataset validation, improved calibration of auxiliary losses, explicit robustness testing, comparison with stronger baselines, and controlled experiments against reference quantum-inspired models. These steps are necessary before making stronger claims about generality, operational deployment, or hardware transferability.

XI. CONCLUSION

A. SUMMARY OF CONTRIBUTIONS

This paper examined Quantum Machine Learning for Time Series Forecasting (QML-TSF) through two complementary components: a systematic review of the literature and a bounded proof-of-concept study of a classical neuro-symbolic Cellular Neural Network (CeNN) for quantum-inspired functional emulation.

The study makes three main contributions. First, it provides a PRISMA-compliant systematic review of 45 QML-TSF studies published or made publicly available between 2014 and 2025 [17]. The review organizes the literature according to algorithmic family, encoding strategy, execution modality, evaluation protocol, and reproducibility practice. It shows that QML-TSF remains a heterogeneous field in which reported gains are often difficult to compare because of inconsistent datasets, horizons, baselines, metrics, execution backends, and reporting standards.

Second, the paper clarifies the distinction between physical quantum execution, classical quantum simulation, and quantum-inspired functional emulation. In this work, functional emulation refers to the classical approximation of selected forecasting-relevant behaviors, including output distributions, spectral structure, temporal correlations, observable-level responses, and predictive performance. It does not imply quantum-state equivalence, genuine entanglement reproduction, physical quantum computation, or quantum computational advantage.

Third, the paper introduces a bounded CeNN-based functional-emulation framework motivated by the scalability and reproducibility gaps identified in the review. The framework uses a locally connected CeNN architecture [16], [44] and a QELM-style reference model [23], [24] to study whether selected quantum-inspired forecasting behaviors can be approximated under a classical, reproducible, and computationally tractable protocol.

B. MAIN FINDINGS

The systematic review indicates that QML-TSF is not yet mature enough to support broad claims of general superiority over classical forecasting models. Many studies report promising results, but the field remains limited by small-scale simulation, NISQ-era constraints, heterogeneous benchmarks, incomplete baseline comparisons, limited statistical testing, and insufficient reproducibility reporting [12], [14], [15].

The proof-of-concept benchmark strengthens the empirical basis of the paper by evaluating the proposed framework on four datasets: energy load, AAPL log-returns, Jena Climate, and Mackey–Glass. The protocol uses up to 20 000 chronological observations per dataset, a 70/15/15 chronological train-validation-test split, a 48-step input window, a 24-step forecasting horizon, and five random seeds. Forecasting performance is evaluated primarily with MASE [49], supported by MAE, RMSE, and R^2.

The empirical results support a bounded conclusion. CeNN_NS achieves competitive forecasting performance against strong neural baselines, but it does not uniformly outperform MLP or LSTM across all datasets. It ranks third on Energy, Jena Climate, and Mackey–Glass, and fourth on AAPL log-returns under the selected protocol. Functional emulation is measurable but domain-dependent: it is strongest on the Mackey–Glass chaotic benchmark and weaker on energy and weather series. The AAPL log-return case is treated cautiously because some naive and QELM outputs become nearly degenerate under the selected transformation.

The scalability analysis supports near-linear architectural growth in the number of CeNN parameters under bounded local-connectivity assumptions. However, training time and inference latency remain sensitive to batching, hardware allocation, software versions, and cloud execution conditions. Therefore, the scalability claim is architectural and functional, not a hardware-independent runtime guarantee.

C. LIMITATIONS

Several limitations remain. First, the benchmark covers four datasets and a single 24-step forecasting horizon. Broader validation is required on additional domains, multivariate series, longer horizons, noisy settings, and regime-shift scenarios.

Second, the emulation results depend on the quality of the QELM reference model. When the teacher is weak or poorly calibrated, alignment with that teacher may not improve forecasting accuracy. This is why the framework includes teacher-quality control, but stronger QELM teachers, ensembles of quantum-inspired references, and adaptive loss weighting should be explored in future work.

Third, the ablation results show that neuro-symbolic and emulation-oriented losses should not be interpreted as universal accuracy boosters. Their main role is to formalize and measure the trade-off between direct forecasting performance and functional alignment with a quantum-inspired reference model.

Fourth, the proposed framework has not been validated on physical quantum hardware. It does not reproduce quantum-state evolution, genuine entanglement, or phase-coherent interference. It should therefore not be interpreted as evidence of physical quantum advantage or as a substitute for QPU-based validation.

D. FUTURE WORK

Future work should proceed in five directions. First, QML-TSF studies need standardized benchmarks with shared datasets, chronological splits, forecasting horizons, preprocessing rules, metrics, and strong baseline protocols. Second, future comparisons should include modern classical forecasting models such as LSTM, Transformer-type models, N-BEATS, Autoformer-like architectures, and domain-specific baselines [3], [4], [54], [55].

Third, reproducibility packages should become standard practice. Code, dataset versions, preprocessing scripts, random seeds, configuration files, environment metadata, generated figures, and output tables should be released whenever licensing and ethical constraints allow. Fourth, hardware-aware evaluation should compare state-vector simulation, noisy simulation, quantum-inspired functional emulation, and physical QPU execution under matched protocols. Fifth, the theory of functional emulation should be strengthened by deriving tighter links between distributional alignment, spectral similarity, temporal-correlation preservation, and forecasting error.

E. FINAL SYNTHESIS

Overall, this paper argues for a disciplined interpretation of QML-TSF. Quantum and quantum-inspired models should not be presented as immediate replacements for classical forecasting systems or as evidence of quantum advantage without controlled, reproducible, and hardware-aware validation. They should instead be evaluated through fair baselines, explicit execution modalities, transparent preprocessing, statistical reporting, and cautious claims.

Within this perspective, CeNN-based neuro-symbolic functional emulation is a promising but bounded research direction. The proposed framework provides a scalable classical pathway for studying selected QELM-related forecasting behaviors, while the empirical results show both its promise and its limits. The contribution should therefore be understood as a reproducible methodological step toward scalable quantum-inspired time-series forecasting, not as a final solution, a universal forecasting architecture, or evidence of physical quantum computational advantage.

DATA AND CODE AVAILABILITY

In alignment with the reproducibility principles encouraged by IEEE Access and in response to the reproducibility limitations identified in the systematic review, we provide a public implementation package for the proposed neuro-symbolic CeNN proof of concept. The repository is intended to support transparent inspection, controlled replication, and further extension of the experiments reported in this paper. It should be understood as a reproducibility-oriented research package rather than as a production-ready software release.

The source code, reproducibility notebook, configuration files, generated outputs, result tables, figures, metadata, and supporting documentation are publicly available in the following GitHub repository:

`GitHub repository for reproducibility package`

The full URL is embedded in the hyperlink above and is also provided in the supplementary submission metadata.

The repository contains the following main components:

- `README.md`: a reviewer-oriented description of the repository, including the scientific scope, experimental protocol, datasets, models, main results, reproducibility instructions, output structure, limitations, and citation information.
- `main_reproducibility_notebook.ipynb`: the main notebook used to reproduce the proof-of-concept benchmark. The notebook loads or prepares the real-world time-series datasets and generates the deterministic Mackey–Glass benchmark internally.
- `requirements.txt`: the dependency file documenting the main software packages required for controlled replication outside the original notebook environment.
- `outputs/`: the output directory containing generated CSV files, figures, metadata, and result summaries.
- `figures/`: the directory containing the exported figures used in the manuscript and supplementary material.

The implementation is designed to reproduce the proof-of-concept benchmark reported in this manuscript. The benchmark uses four datasets covering energy load forecasting, financial log-return forecasting, weather temperature forecasting, and deterministic chaotic dynamics. It follows a fixed protocol based on up to 20 000 chronological observations per dataset, a 70/15/15 chronological train-validation-test split, a 48-step input window, a 24-step forecasting horizon, and five random seeds for trainable models:

$$\mathcal{S} = \{42, 123, 2024, 7, 99\}.$$

The four datasets are handled as follows:

- **Energy**: real-world electricity load time series used for energy load forecasting.
- **AAPL**: financial time series transformed into log-returns before forecasting. The log-return transformation is used to avoid interpreting near-persistence behavior in raw prices as strong forecasting evidence.
- **Jena Climate**: weather time series used for temperature forecasting.

- **Mackey–Glass**: deterministic chaotic series generated within the notebook and used as a controlled nonlinear benchmark.

No synthetic fallback dataset was used in the reported benchmark. Although the notebook may contain fallback or diagnostic logic for pipeline testing, only the declared benchmark datasets contribute to the results reported in the manuscript.

The key reproducibility files include:

- `summary_full.csv`;
- `benchmark_full.csv`;
- `emulation_summary_full.csv`;
- `emulation_fidelity_full.csv`;
- `ablation_full.csv`;
- `scalability_full.csv`;
- `linear_fit_full.csv`;
- `statistical_comparison_full.csv`;
- `metadata_full.json`;
- exported figures for forecasting performance, functional emulation, ablation, scalability, teacher selection, and framework architecture.

Because cloud and GPU environments may introduce small numerical variations, the repository is not intended to guarantee bitwise-identical reproduction across all systems. Instead, it provides the information required for controlled replication within reasonable numerical tolerance. This includes dataset handling instructions, preprocessing rules, chronological splits, random seeds, configuration files, software dependencies, generated outputs, figures, and metadata.

By releasing this reproducibility package, we aim to address one of the main weaknesses identified in the reviewed QML-TSF literature: incomplete reporting of code, seeds, preprocessing details, software versions, generated outputs, and experimental configurations. The repository is intended to help reviewers and researchers inspect the implementation, verify the reported proof-of-concept experiments, and extend the CeNN framework to additional datasets and forecasting settings.

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APPENDIX A SIMULATION ASSUMPTIONS AND REPRODUCIBILITY NOTES

This appendix summarizes the assumptions used for the illustrative memory-complexity comparison between state-vector simulation and CeNN-based functional emulation. The purpose of this appendix is limited: it documents the scaling assumptions used in the manuscript and clarifies the reproducibility scope of the proof-of-concept experiments.

A. MEMORY-COMPLEXITY ASSUMPTIONS

The comparison between quantum state-vector simulation and CeNN-based functional emulation relies on the following simplified assumptions.

- **State-vector simulation:** an n-qubit pure quantum state requires 2^n complex amplitudes. Under double-precision complex representation, each amplitude requires approximately 16 bytes. This leads to exponential memory growth in n [14], [15].
- **CeNN functional emulator:** the CeNN representation consists of N classical real-valued cells. Each cell stores its state and a bounded number of local template parameters. Under a fixed neighborhood size k, the memory footprint scales as $\mathcal{O}(N)$.
- **Illustrative threshold:** for comparison, a classical compute node with 128 GB of RAM is used as a reference memory threshold.

This comparison is intended only to illustrate the difference between full quantum-state representation and classical functional emulation. It does not imply that the CeNN representation is physically equivalent to an n-qubit quantum state. A CeNN cell is a classical dynamical unit and should not be interpreted as a qubit.

B. REPRODUCIBILITY STATEMENT

The proof-of-concept experiments and generated figures can be reproduced using the notebook, configuration files, exported tables, and generated outputs provided in the public repository. Because cloud GPU environments may vary across sessions, small numerical differences may occur. The goal is controlled replication within reasonable numerical tolerance rather than bitwise-identical reproduction.

APPENDIX B RISK-OF-BIAS ASSESSMENT DETAILS

This appendix provides additional details on the risk-of-bias and quality-assessment procedure used for the studies included in the systematic review. The assessment supports the interpretation of the synthesized findings and highlights recurring methodological limitations in QML-TSF research. The rubric is inspired by the logic of structured risk-of-bias assessment, while being adapted to the specific requirements of QML-TSF studies [17], [53].

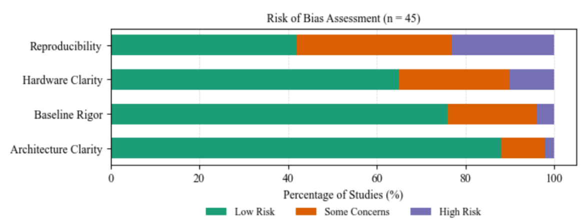

FIGURE 7. Risk-of-bias summary across the included QML-TSF corpus according to the four quality domains D1–D4.

A. QUALITY DOMAINS

The assessment was structured around four domains adapted to the QML-TSF context:

TABLE 17. Risk-of-bias assessment results across the included studies.

Quality domain	Low risk	Some concerns	High risk
D1: Architectural transparency	88% (40)	10% (4)	2% (1)
D2: Baseline rigor	76% (34)	20% (9)	4% (2)
D3: Platform and noise clarity	65% (29)	25% (11)	10% (5)
D4: Reproducibility	42% (19)	35% (16)	23% (10)

Numbers in parentheses indicate the count of studies in each category. Inter-rater agreement: Cohen's $\kappa = 0.85$.

1) **D1: Architectural transparency** assesses whether the study sufficiently describes the QML or quantum-inspired model, including circuit design, encoding strategy, qubit count where applicable, circuit depth, parameter count, and model configuration.
2) **D2: Baseline rigor** assesses whether the study compares the proposed model against appropriate and sufficiently tuned classical or non-quantum baselines.
3) **D3: Platform and noise clarity** assesses whether the study explicitly reports whether results were obtained through simulation, emulation, or physical hardware, together with software backend, hardware details, shot count, or noise assumptions where applicable.
4) **D4: Reproducibility** assesses whether code, datasets, preprocessing details, random seeds, train-validation-test splits, and configuration files are available or sufficiently described.

B. KEY OBSERVATION

The assessment indicates that architectural transparency is generally stronger than reproducibility. In particular, D4 remains the weakest domain, with only 42% of the included studies rated as low risk. This finding supports the need for reproducibility packages, standardized benchmarks, and clearer reporting of experimental configurations in future QML-TSF research.

APPENDIX C CLASSICAL FORECASTING BASELINES

This appendix briefly summarizes the classical forecasting model families used as reference points throughout the manuscript. The purpose is not to provide a full tutorial on classical forecasting, but to clarify why strong classical baselines remain essential when evaluating QML-TSF claims.

A. CLASSICAL STATISTICAL MODELS

Statistical models, such as ARIMA and seasonal variants, rely on temporal dependence and stationarity assumptions. They remain useful because of their interpretability and strong theoretical grounding, but they may be limited under strongly non-linear, non-stationary, or chaotic temporal dynamics [1], [2].

B. MACHINE LEARNING MODELS

Machine learning models such as support vector regression, random forests, and gradient-boosting methods can capture non-linear relationships, but their performance often depends on lag selection, rolling statistics, calendar variables, transformations, and other domain-specific features.

C. DEEP LEARNING MODELS

Deep learning architectures, including MLPs, recurrent neural networks, LSTMs, and Transformer-type models, can learn temporal representations directly from data [3], [4]. However, they may require substantial data, careful regularization, hyperparameter tuning, and computational resources. Their performance may also degrade under distribution shift, data scarcity, or weakly controlled evaluation protocols.

APPENDIX D PRISMA 2020 CHECKLIST AND METHODOLOGY COMPLIANCE

This appendix summarizes the PRISMA 2020 reporting compliance of the systematic review [17]. The checklist maps the relevant PRISMA items to the corresponding sections of the manuscript, including search strategy, eligibility criteria, study selection, data extraction, risk-of-bias assessment, and synthesis strategy.

The study selection process, eligibility criteria, quality assessment, and synthesis methods are described in Section III. The PRISMA flow diagram is provided in Fig. 1. A completed PRISMA checklist may be provided as supplementary material when required by the journal.

APPENDIX E REFERENCE CLASSIFICATION USED IN THE META-SYNTHESIS

Table 18 classifies the records used to support the systematic review, taxonomy, and meta-synthesis. The table distinguishes empirical QML-TSF studies from benchmarking papers, software-oriented contributions, theoretical foundations, and methodological references. This distinction is important because not all cited records are empirical time-series forecasting studies under the PRISMA eligibility criteria.

TABLE 18. Classification of Cited Records Supporting the QML-TSF Review, Taxonomy, and Meta-Synthesis

Reference	Primary role / method	Domain	RoB / Rel.	Compact note
[10]	QNN/VQC	Energy / TSF	Low	Empirical QML-TSF; clear architecture and baselines
[28]	Review / QSVM context	Multi-domain	Context	Background for QML–classical comparison
[69]	QRC / quantum outcomes	Quantum hardware	Low	Hardware-oriented TS analysis with noise focus
[29]	QNN	Energy / TSF	Some Concerns	Empirical QML-TSF; partial reporting
[70]	QNN / quantum recurrent model	Traffic / TSF	Low	Empirical forecasting; clear model description
[11]	QLSTM	Solar energy	Low	Comparative QLSTM–LSTM forecasting
[35]	QSVM / QML finance	Finance	Some Concerns	Finance QML forecasting; limited baseline detail
[13]	QRNN emulation	Multivariate / chaotic TSF	Low	Density-matrix emulation for TS prediction
[24]	QELM/EQELM	Long-term TSF	Low	Long-horizon quantum-inspired forecasting
[12]	QML benchmark	Multi-domain TSF	Low	Benchmarking study; baseline fairness reference
[19]	QRC	Chaotic series	Some Concerns	QRC for TS prediction; configuration-sensitive
[18]	QRC	Chaotic / dynamical series	Low	QRC with weak/projective measurements
[25]	QRC	Time-series analysis	Some Concerns	Feedback-driven reservoir
[22]	QNN / data re-uploading	Traffic forecasting	Low	Urban traffic QNN forecasting
[43]	QNN	Demand / energy forecasting	Some Concerns	Empirical QNN forecasting; limited validation
[31]	QRC	Chaotic dynamics	Low	QRC for chaotic and extreme-event prediction
[30]	Quantum-kernel LSTM	Climate forecasting	Low	Hybrid quantum-kernel recurrent model
[59]	QRC software	Software / QRC	Context	Software support for reproducible QRC
[58]	Software framework	Hybrid QML	Context	PennyLane hybrid QML framework
[16]	CeNN theory	Dynamical systems	Foundational	Foundational CeNN theory reference
[44]	CeNN applications	Dynamical systems	Foundational	CeNN application background
[45]	Neuro-symbolic AI	AI methodology	Foundational	Neuro-symbolic regularization context
[56]	Neuro-symbolic integration	Cognitive / AI	Context	Symbolic integration and interpretability
[46]	Neuro-symbolic AI survey	AI methodology	Foundational	Survey on neuro-symbolic learning
[71]	QGAN / anomaly detection	Network TS	Some Concerns	Quantum-inspired TS anomaly work
[72]	Quantum-inspired review	Optimization / learning	Context	Background on quantum-inspired learning
[73]	Quantum cognition review	Cognition	Context	Background; not empirical QML-TSF evidence
[9]	QML perspective	QML theory	Foundational	Quantum feature-space perspective
[23]	QELM review	QELM / QML	Context	QELM background; not standalone TSF benchmark
[74]	Quantum learning theory	Game theory	Context	Theory reference; not empirical TSF evidence
[20]	QNN / QML forecasting	Time series	Some Concerns	Empirical QML-TSF; partial reporting
[38]	VQC / hybrid QML	Energy / demand response	Low	Hybrid quantum–classical energy application
[39]	QSVM	Energy forecasting	Some Concerns	Comparative energy forecasting
[40]	Hybrid QML	Wind energy	Some Concerns	Offshore wind power forecasting
[32]	QRC	Finance / volatility	Low	QRC for realized volatility forecasting
[42]	Quantum genetic LVQ	Traffic forecasting	High	Limited reproducibility and baseline detail
[41]	QNN	Wind speed forecasting	Some Concerns	Short-term wind-speed QNN forecasting
[75]	Review / QML finance	Finance	Context	AI/QML stock prediction background
[76]	Explainable AI	Multivariate TS	Context	Counterfactual explanation context
[37]	Quantum deep learning	Energy / oil futures	Some Concerns	QDL for crude-oil futures forecasting
[26]	Quantum graph model	Traffic forecasting	Low	Spatio-temporal quantum graph forecasting
[33]	QRC	Market / food prices	Some Concerns	QRC for market forecasting
[6]	QML review	QML theory	Foundational	Foundational QML reference
[8]	Quantum feature maps	Classification / kernels	Foundational	Quantum feature-map background
[14]	VQA review	Variational algorithms	Foundational	VQA, NISQ, and barren-plateau context

Note: Records marked as *Low*, *Some Concerns*, or *High* correspond to empirical or benchmarking-oriented QML-TSF evidence assessed through the D1–D4 framework. Records marked as *Foundational* or *Context* are supporting theoretical, methodological, software, or background references and are not counted as empirical QML-TSF benchmark studies.

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